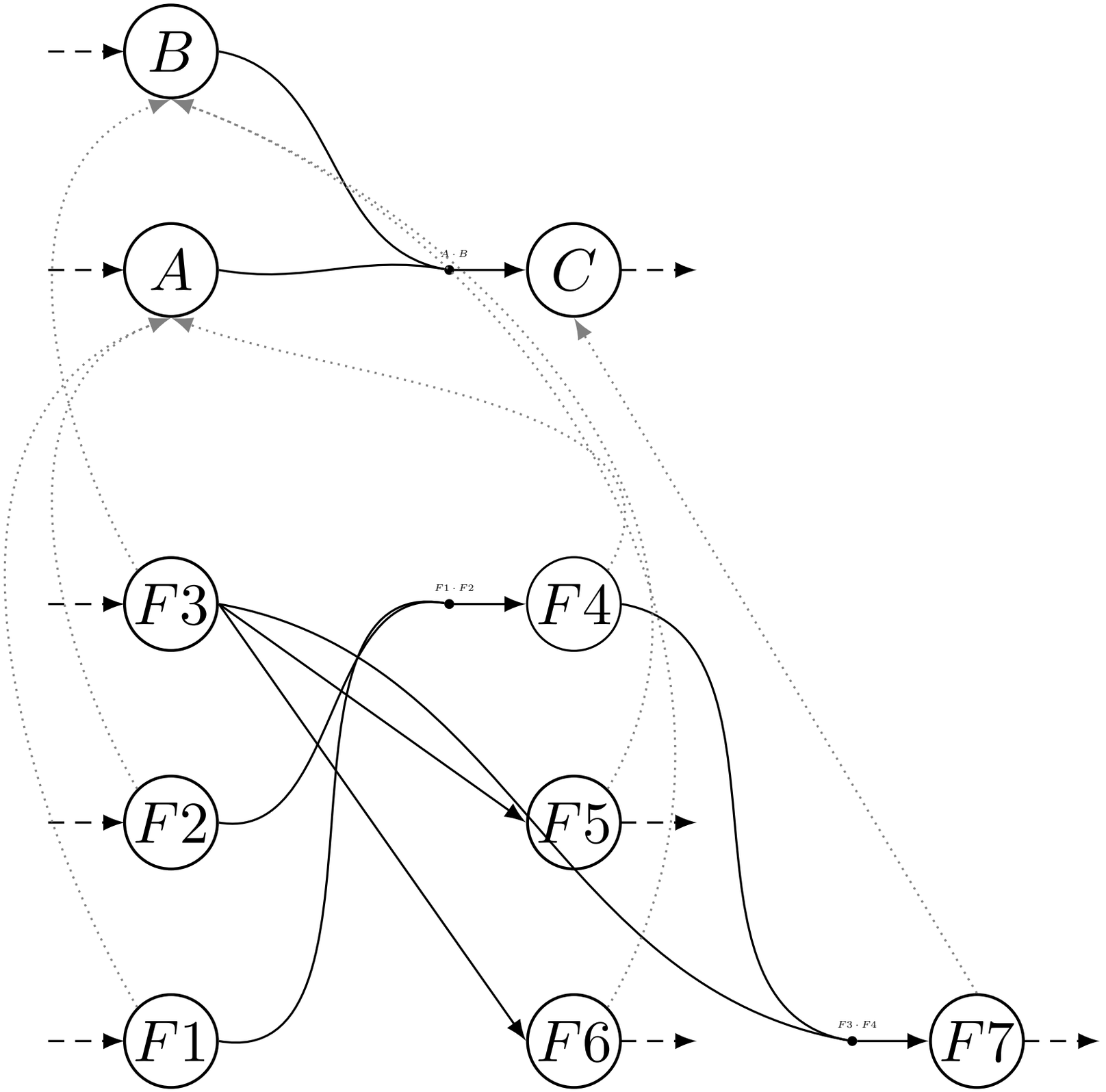
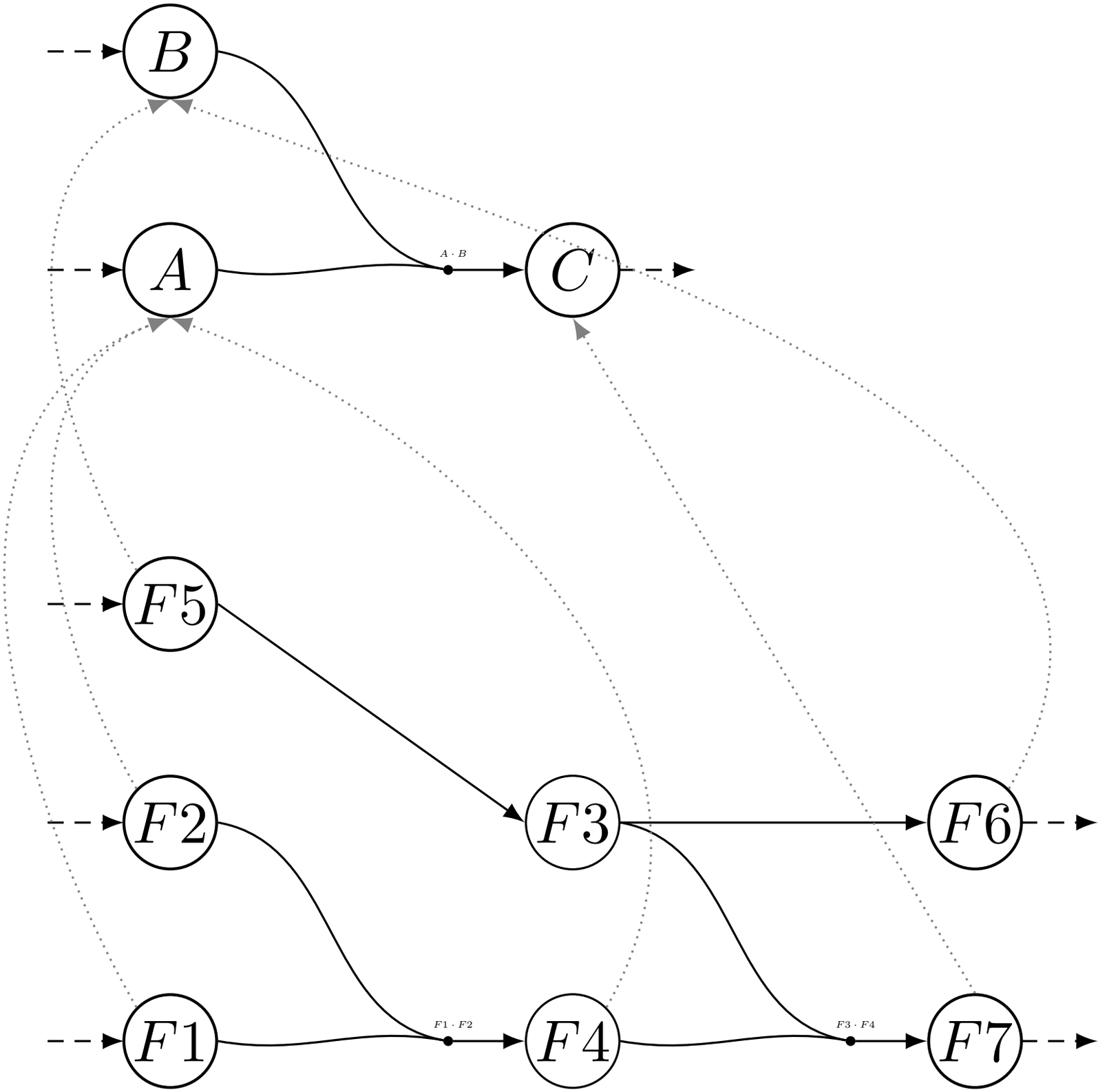
solution no. 1/54



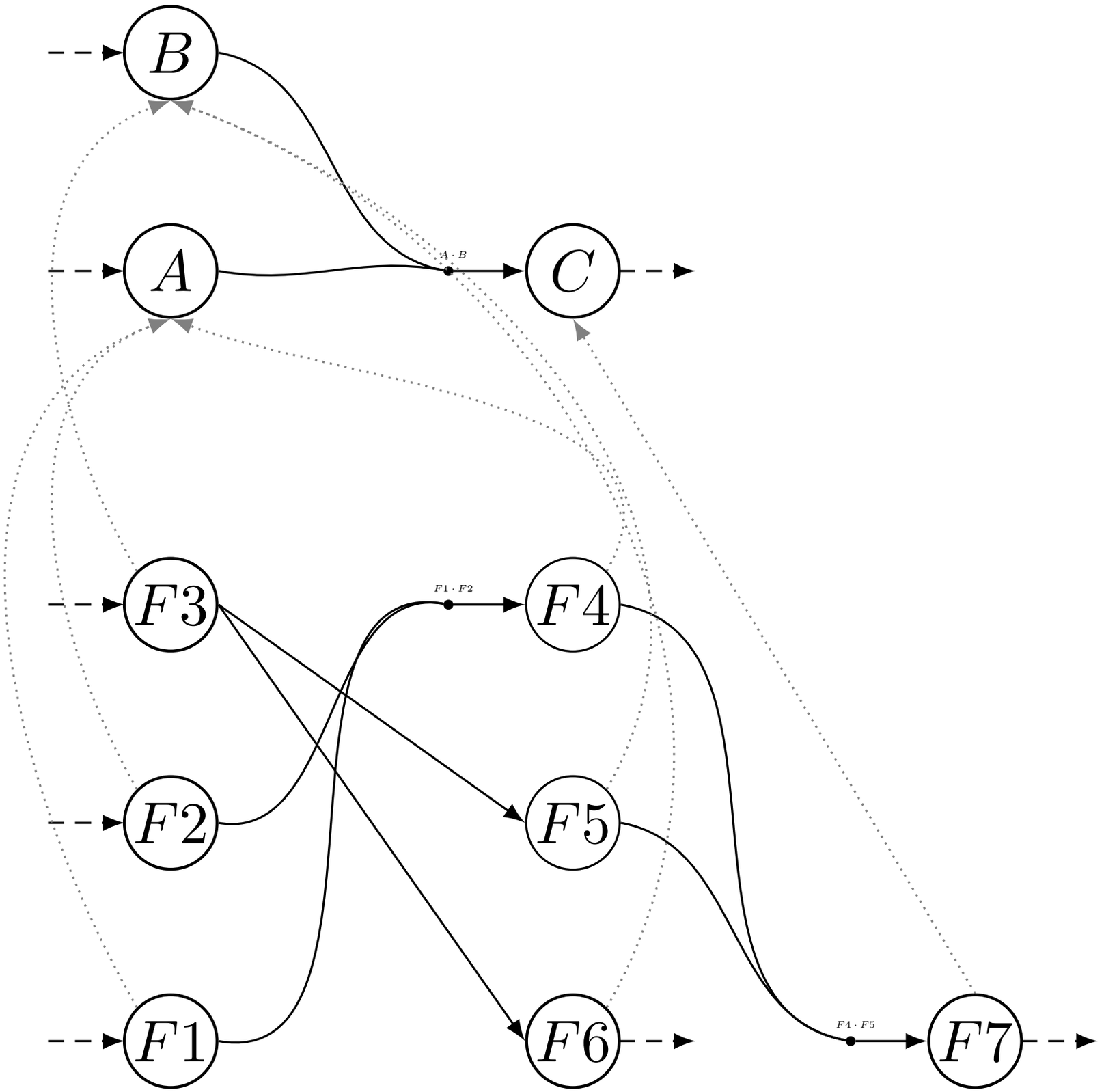
$$(A \cdot B \leftrightarrow C) \cdot (F1 \cdot F2 \leftrightarrow F4) \cdot (F3 \leftrightarrow F5) \cdot (F3 \leftrightarrow F6) \cdot (F3 \cdot F4 \leftrightarrow F7)$$

solution no. 2/54



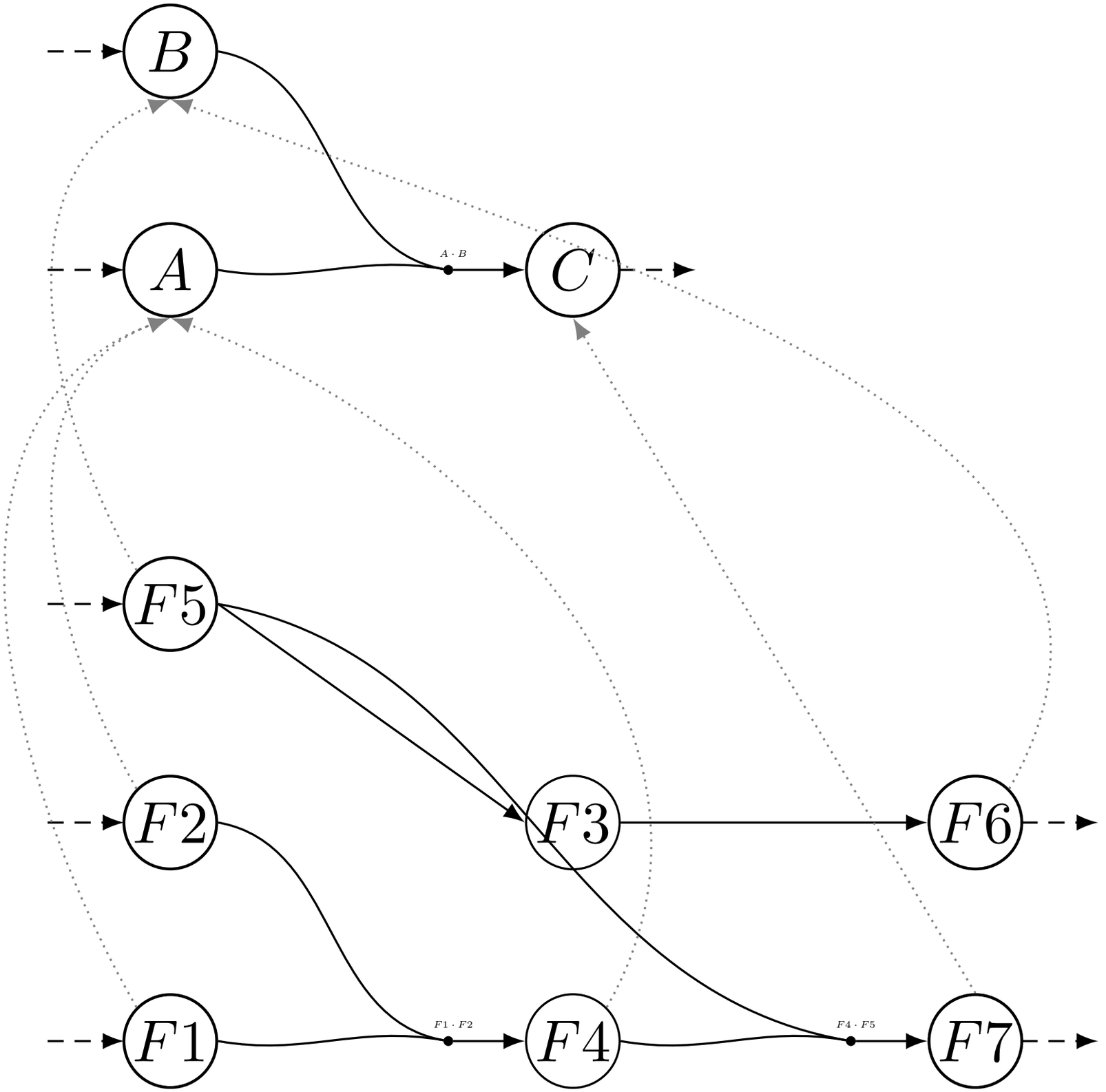
$$(A \cdot B \leftrightarrow C) \cdot (F5 \leftrightarrow F3) \cdot (F1 \cdot F2 \leftrightarrow F4) \cdot (F3 \leftrightarrow F6) \cdot (F3 \cdot F4 \leftrightarrow F7)$$

solution no. 3/54



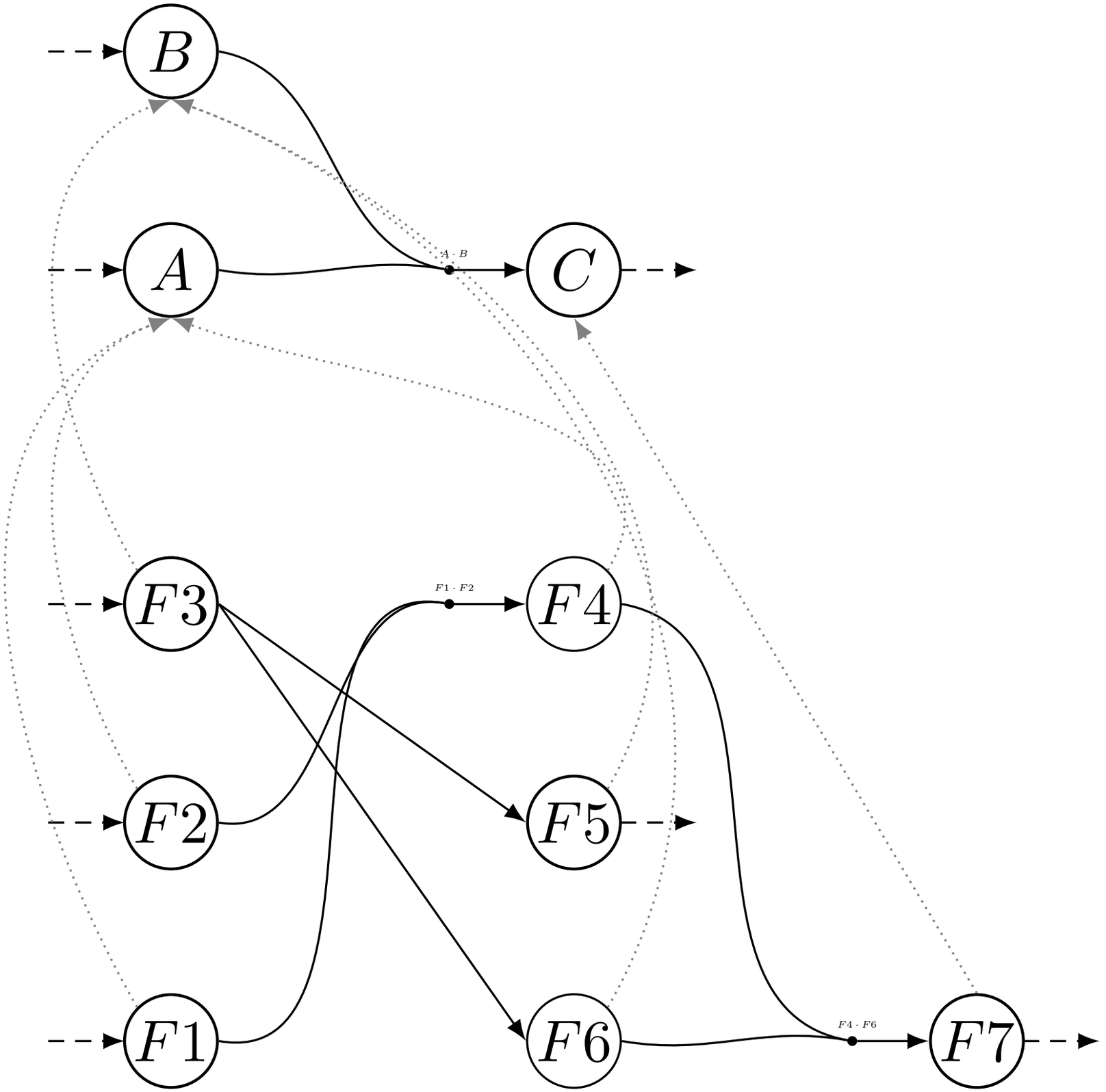
$$(A \cdot B \leftrightarrow C) \cdot (F1 \cdot F2 \leftrightarrow F4) \cdot (F3 \leftrightarrow F5) \cdot (F3 \leftrightarrow F6) \cdot (F4 \cdot F5 \leftrightarrow F7)$$

solution no. 4/54



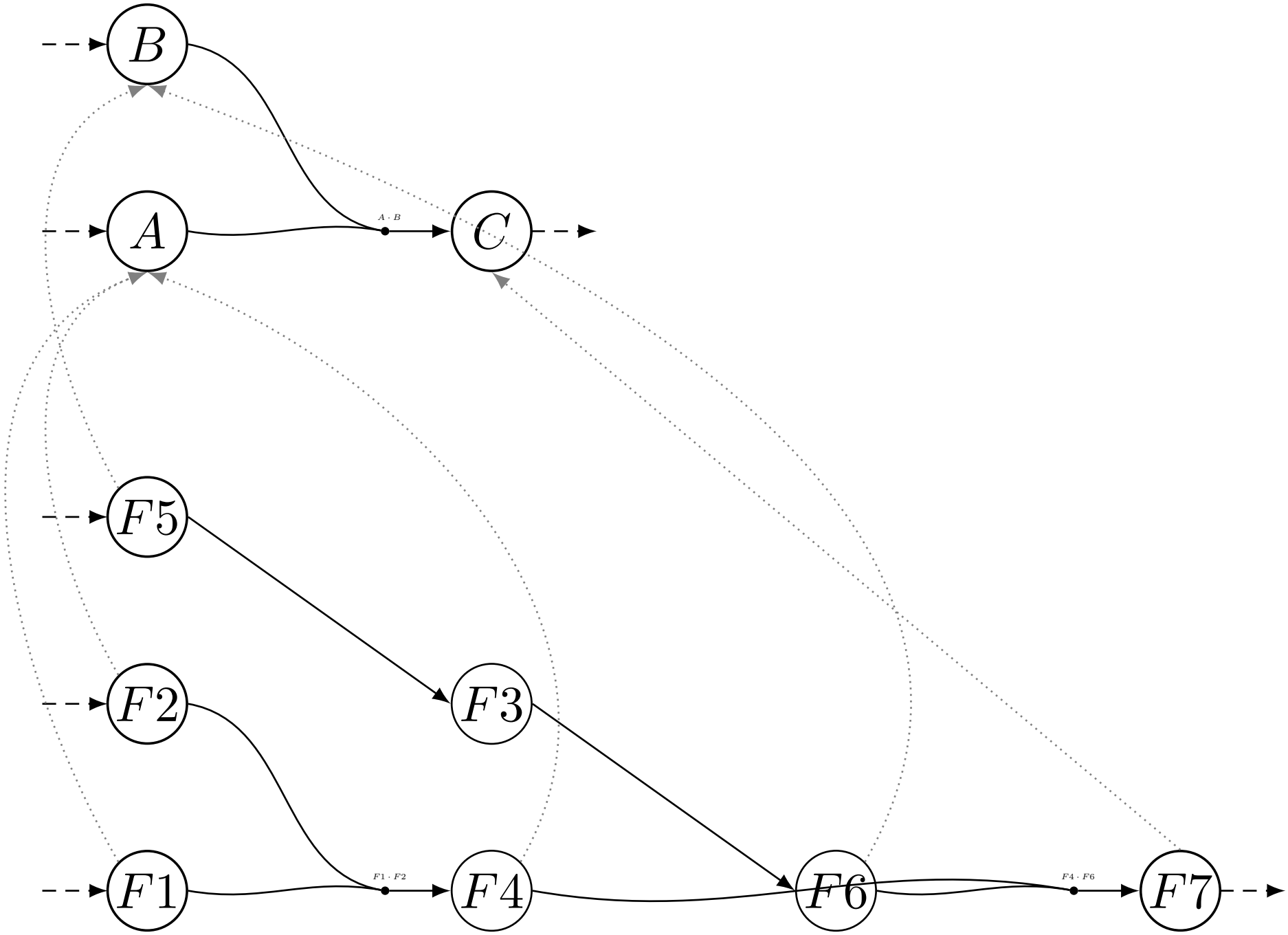
$$(A \cdot B \leftrightarrow C) \cdot (F5 \leftrightarrow F3) \cdot (F1 \cdot F2 \leftrightarrow F4) \cdot (F3 \leftrightarrow F6) \cdot (F4 \cdot F5 \leftrightarrow F7)$$

solution no. 5/54



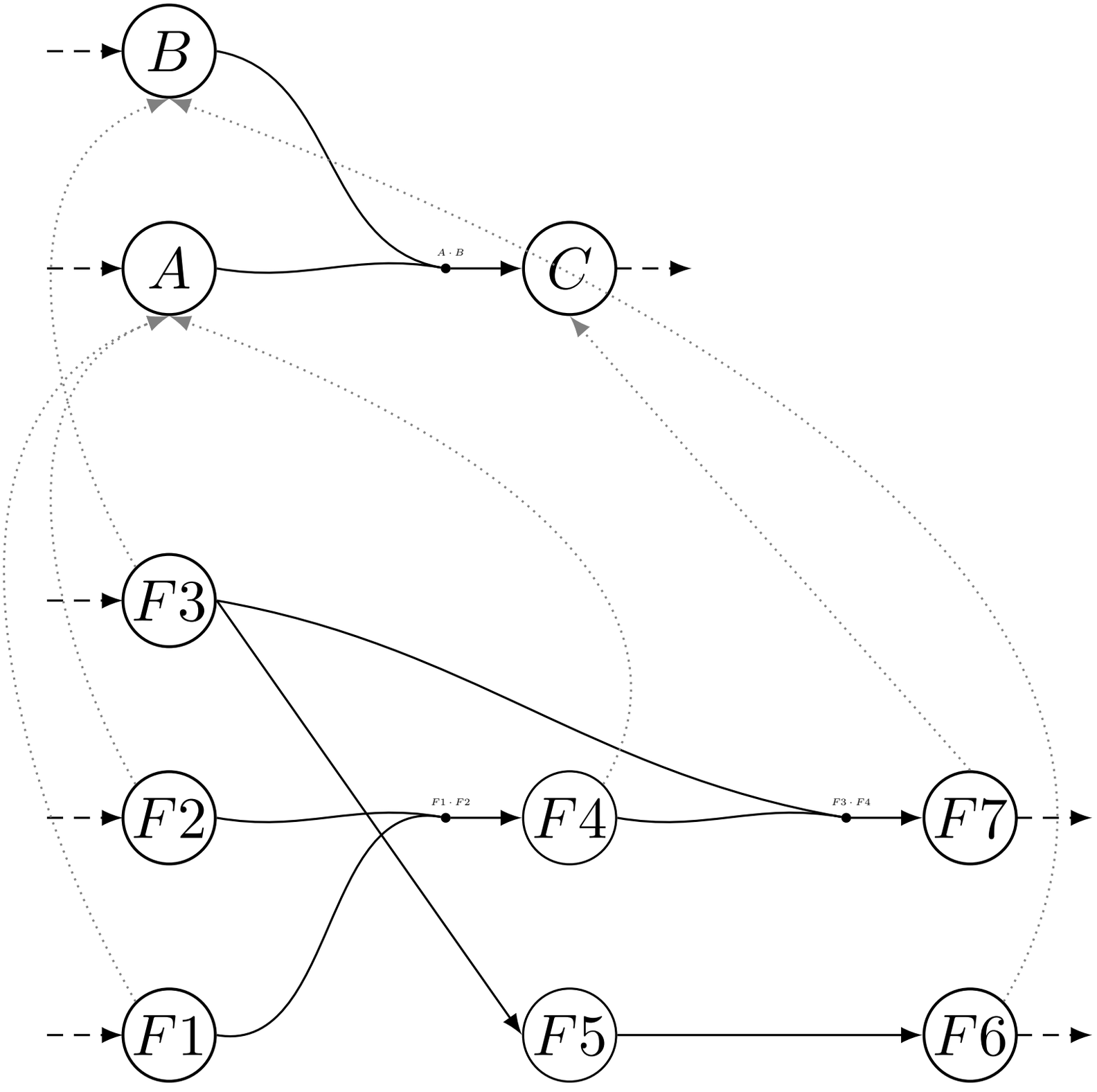
$$(A \cdot B \leftrightarrow C) \cdot (F1 \cdot F2 \leftrightarrow F4) \cdot (F3 \leftrightarrow F5) \cdot (F3 \leftrightarrow F6) \cdot (F4 \cdot F6 \leftrightarrow F7)$$

solution no. 6/54



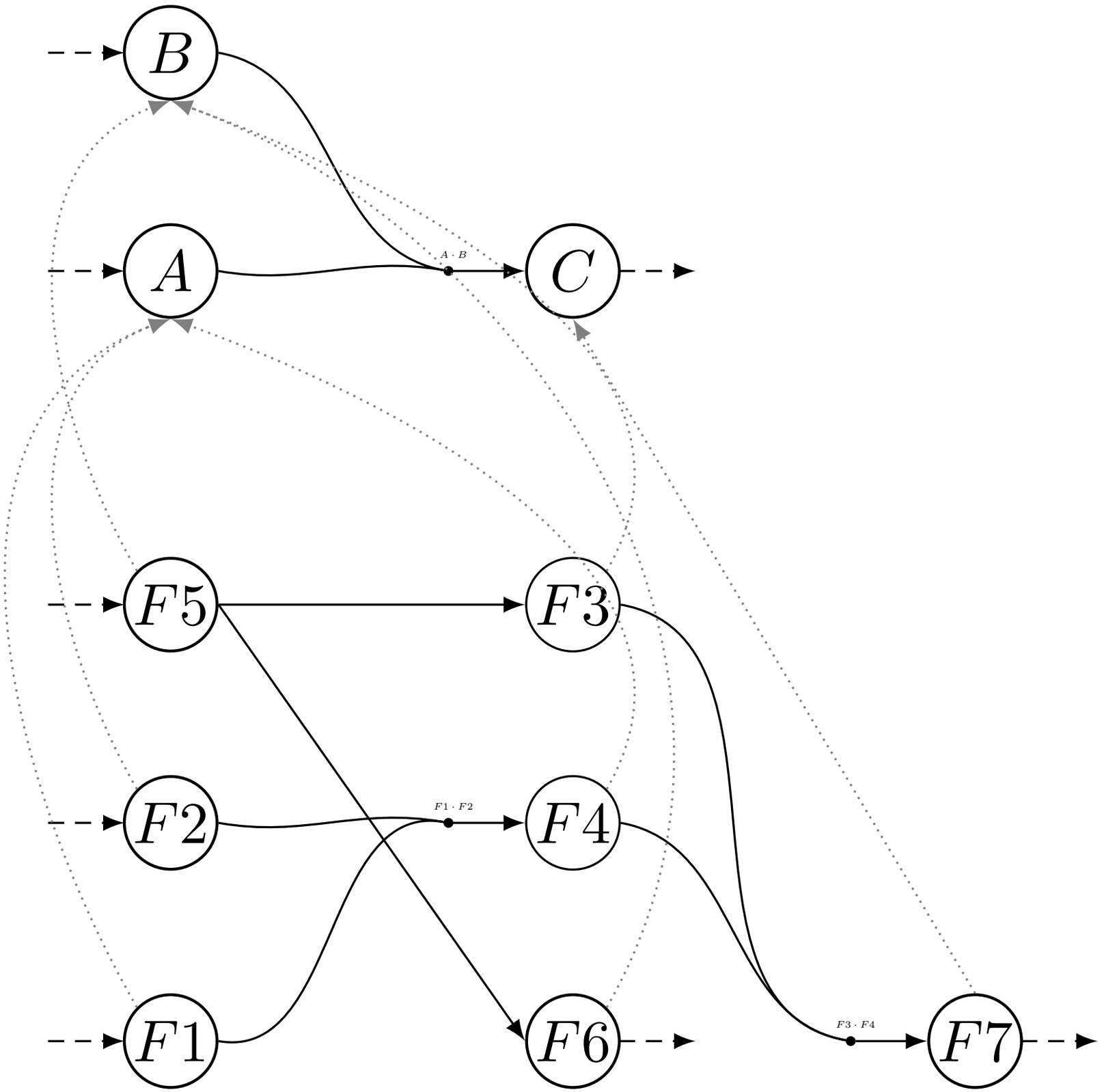
$$(A \cdot B \leftrightarrow C) \cdot (F5 \leftrightarrow F3) \cdot (F1 \cdot F2 \leftrightarrow F4) \cdot (F3 \leftrightarrow F6) \cdot (F4 \cdot F6 \leftrightarrow F7)$$

solution no. 7/54



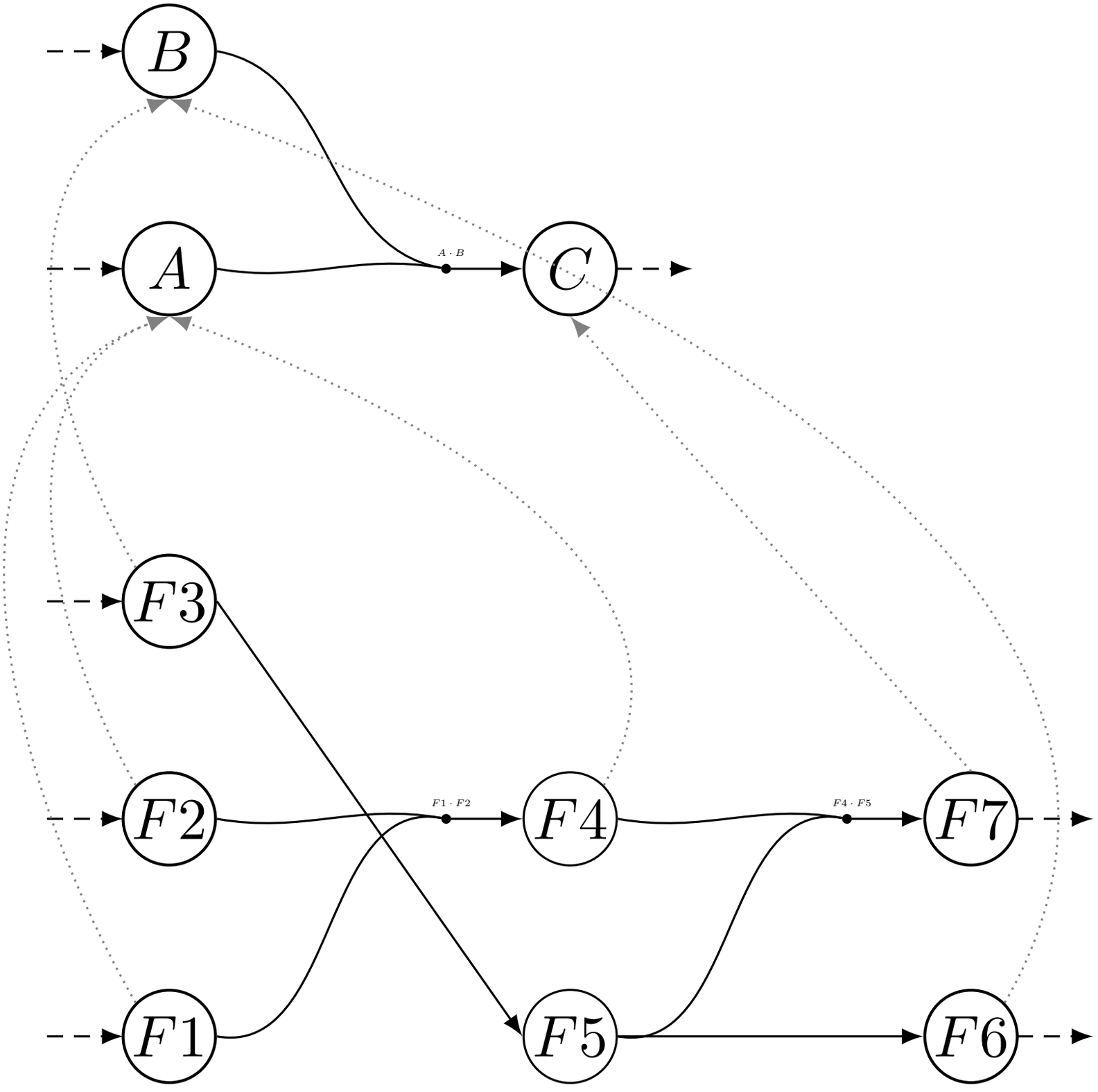
$$(A \cdot B \leftrightarrow C) \cdot (F1 \cdot F2 \leftrightarrow F4) \cdot (F3 \leftrightarrow F5) \cdot (F5 \leftrightarrow F6) \cdot (F3 \cdot F4 \leftrightarrow F7)$$

solution no. 8/54



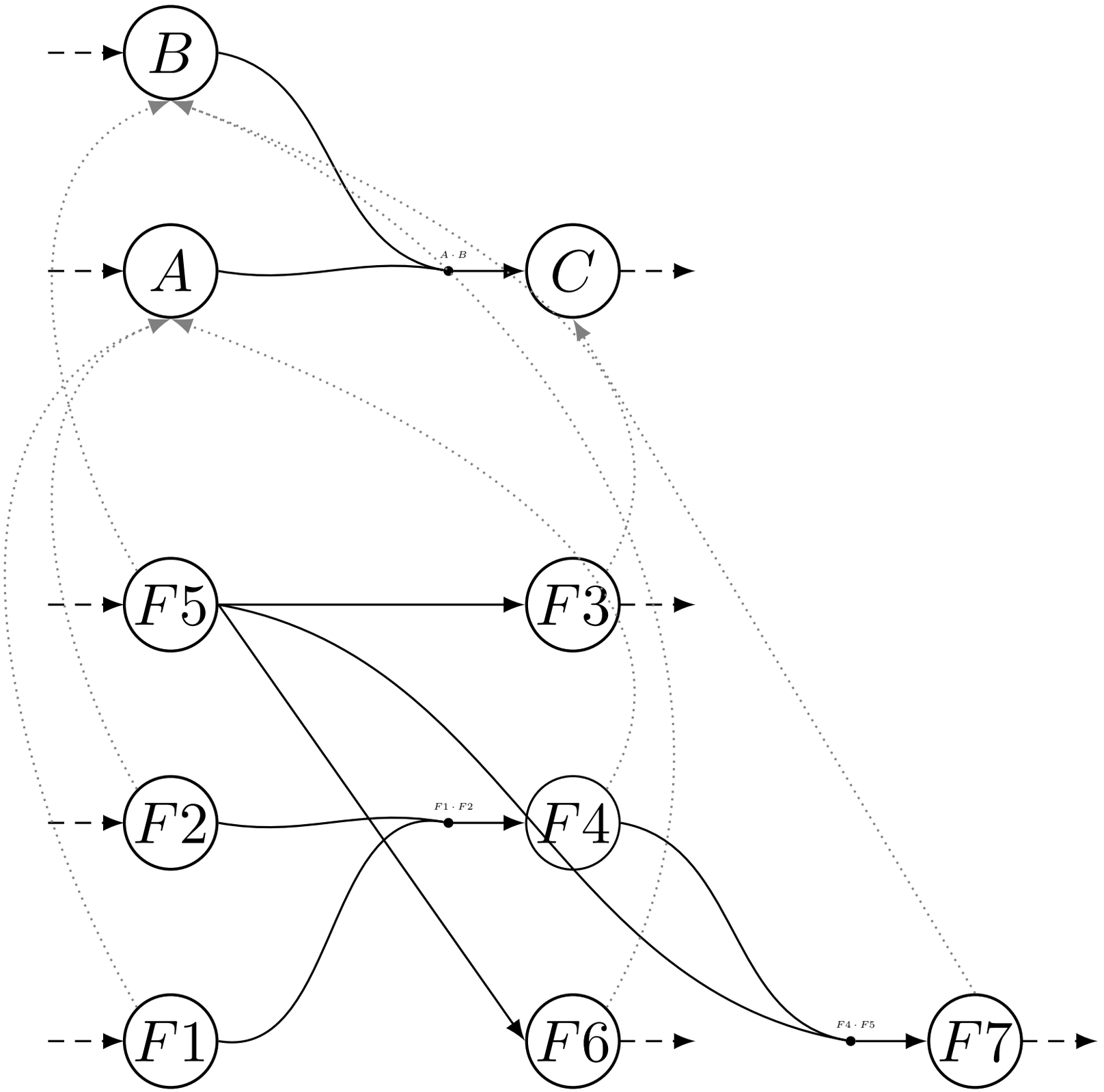
$$(A \cdot B \leftrightarrow C) \cdot (F5 \leftrightarrow F3) \cdot (F1 \cdot F2 \leftrightarrow F4) \cdot (F5 \leftrightarrow F6) \cdot (F3 \cdot F4 \leftrightarrow F7)$$

solution no. 9/54



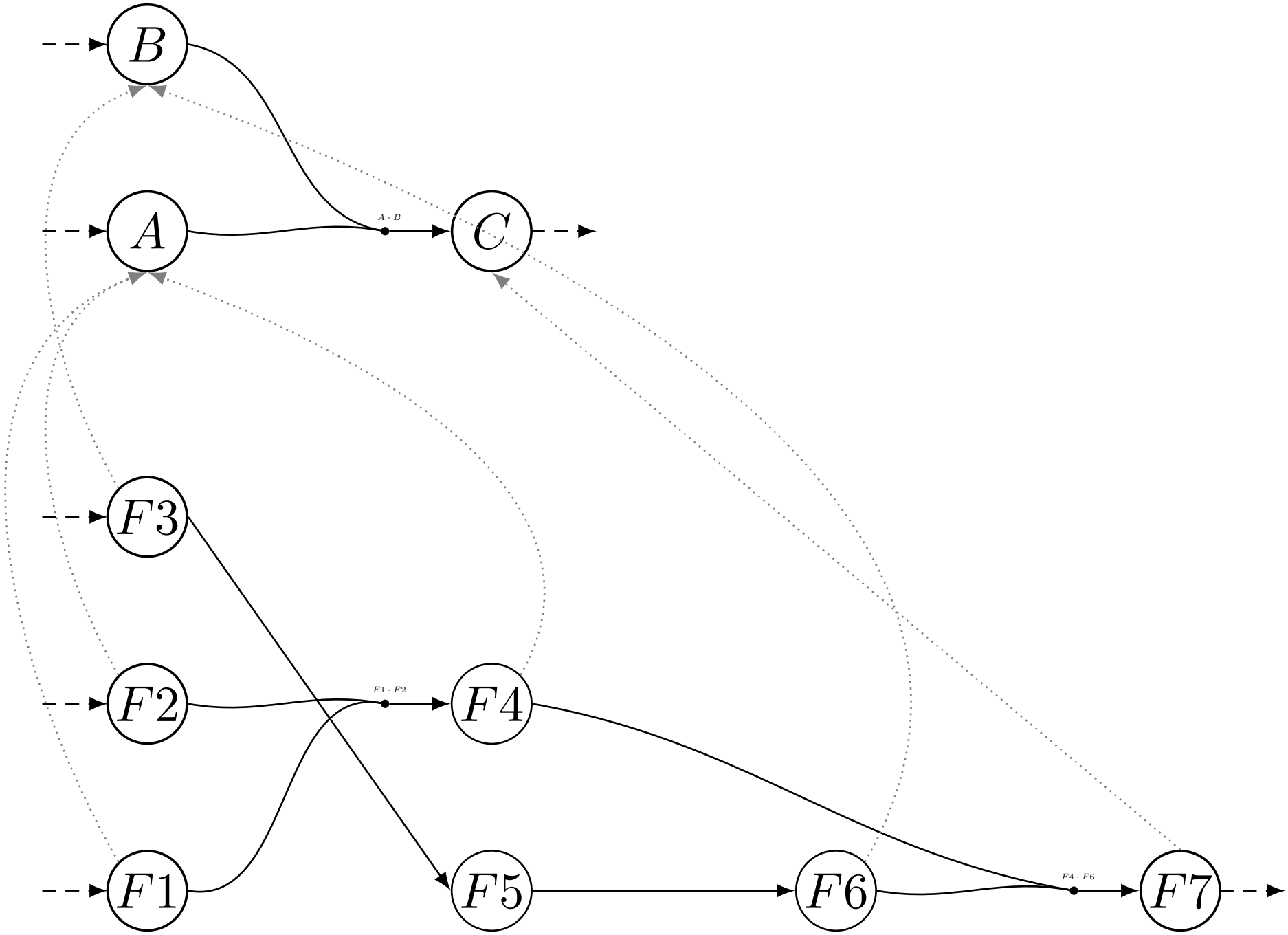
$$(A \cdot B \leftrightarrow C) \cdot (F1 \cdot F2 \leftrightarrow F4) \cdot (F3 \leftrightarrow F5) \cdot (F5 \leftrightarrow F6) \cdot (F4 \cdot F5 \leftrightarrow F7)$$

solution no. 10/54



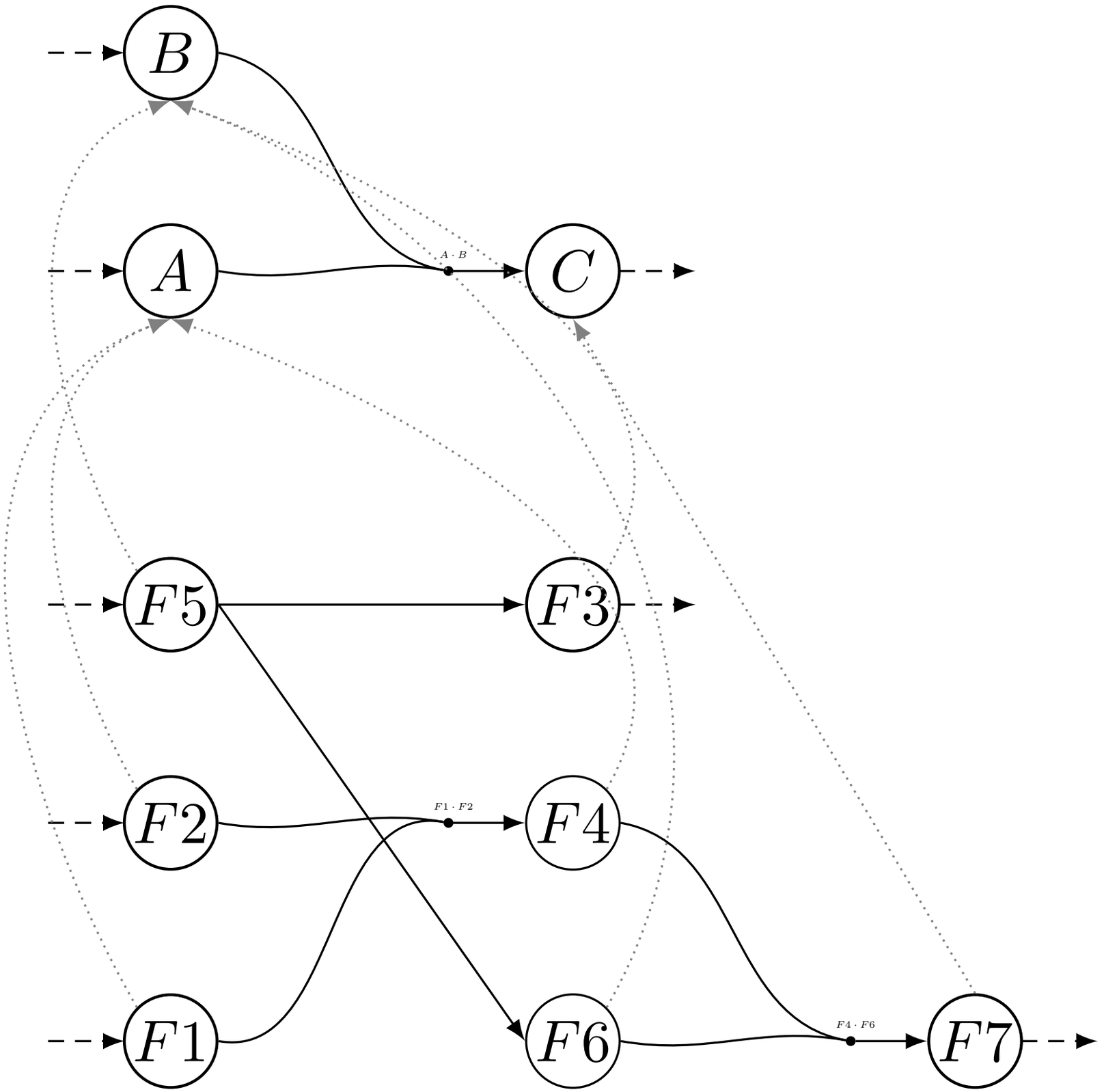
$$(A \cdot B \leftrightarrow C) \cdot (F5 \leftrightarrow F3) \cdot (F1 \cdot F2 \leftrightarrow F4) \cdot (F5 \leftrightarrow F6) \cdot (F4 \cdot F5 \leftrightarrow F7)$$

solution no. 11/54



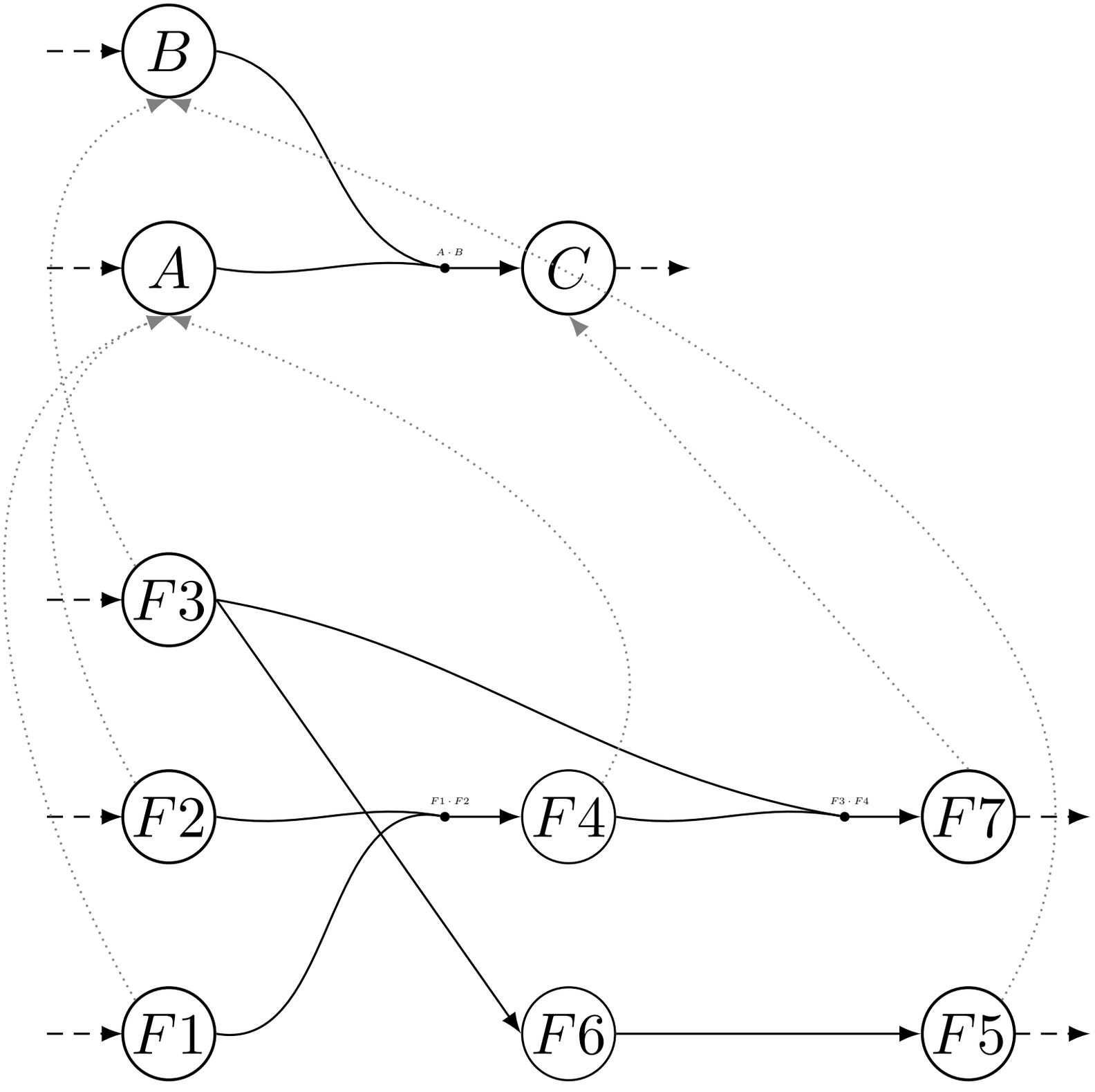
$$(A \cdot B \leftrightarrow C) \cdot (F1 \cdot F2 \leftrightarrow F4) \cdot (F3 \leftrightarrow F5) \cdot (F5 \leftrightarrow F6) \cdot (F4 \cdot F6 \leftrightarrow F7)$$

solution no. 12/54



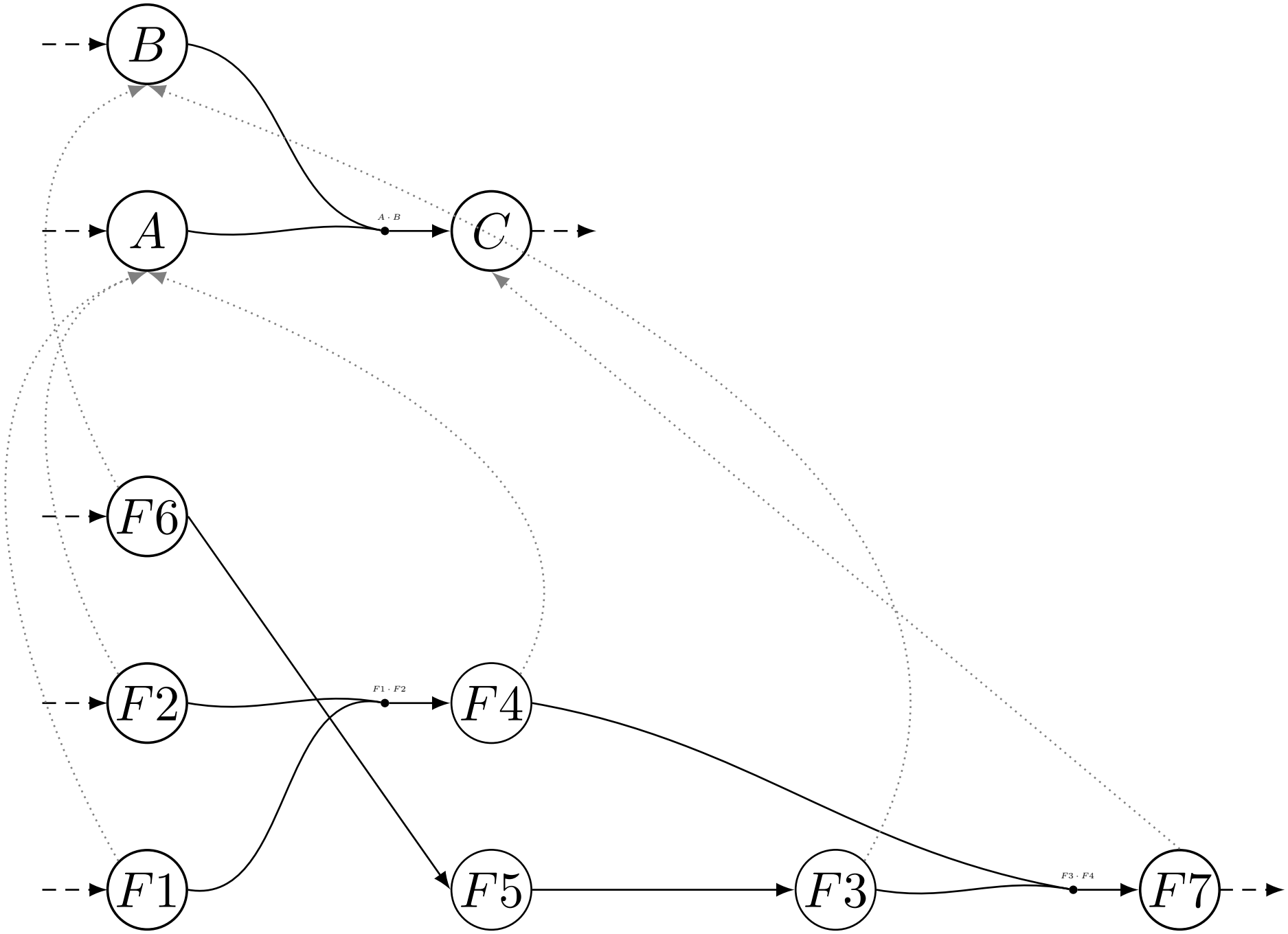
$$(A \cdot B \leftrightarrow C) \cdot (F5 \leftrightarrow F3) \cdot (F1 \cdot F2 \leftrightarrow F4) \cdot (F5 \leftrightarrow F6) \cdot (F4 \cdot F6 \leftrightarrow F7)$$

solution no. 13/54



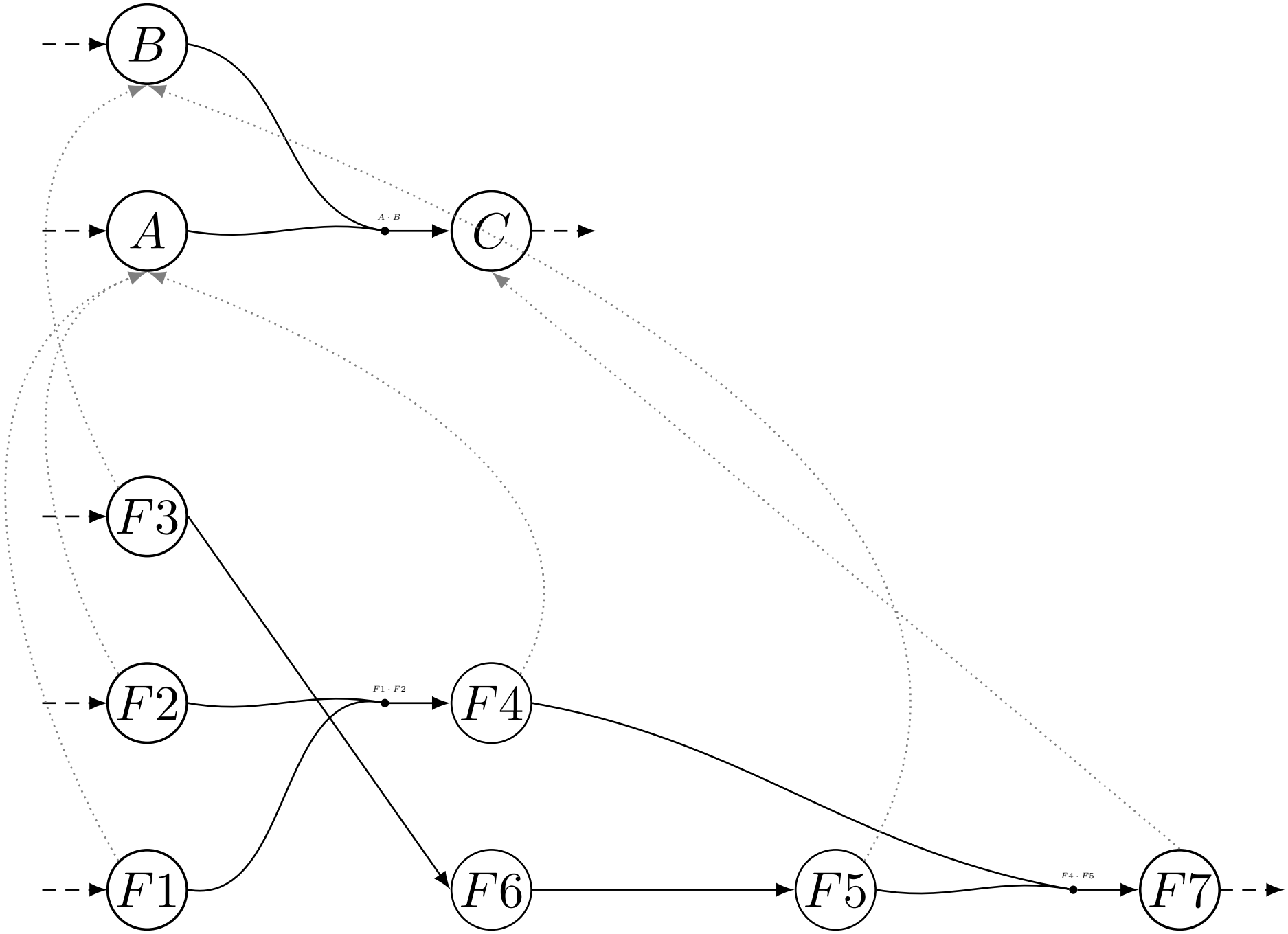
$$(A \cdot B \leftrightarrow C) \cdot (F1 \cdot F2 \leftrightarrow F4) \cdot (F6 \leftrightarrow F5) \cdot (F3 \leftrightarrow F6) \cdot (F3 \cdot F4 \leftrightarrow F7)$$

solution no. 14/54



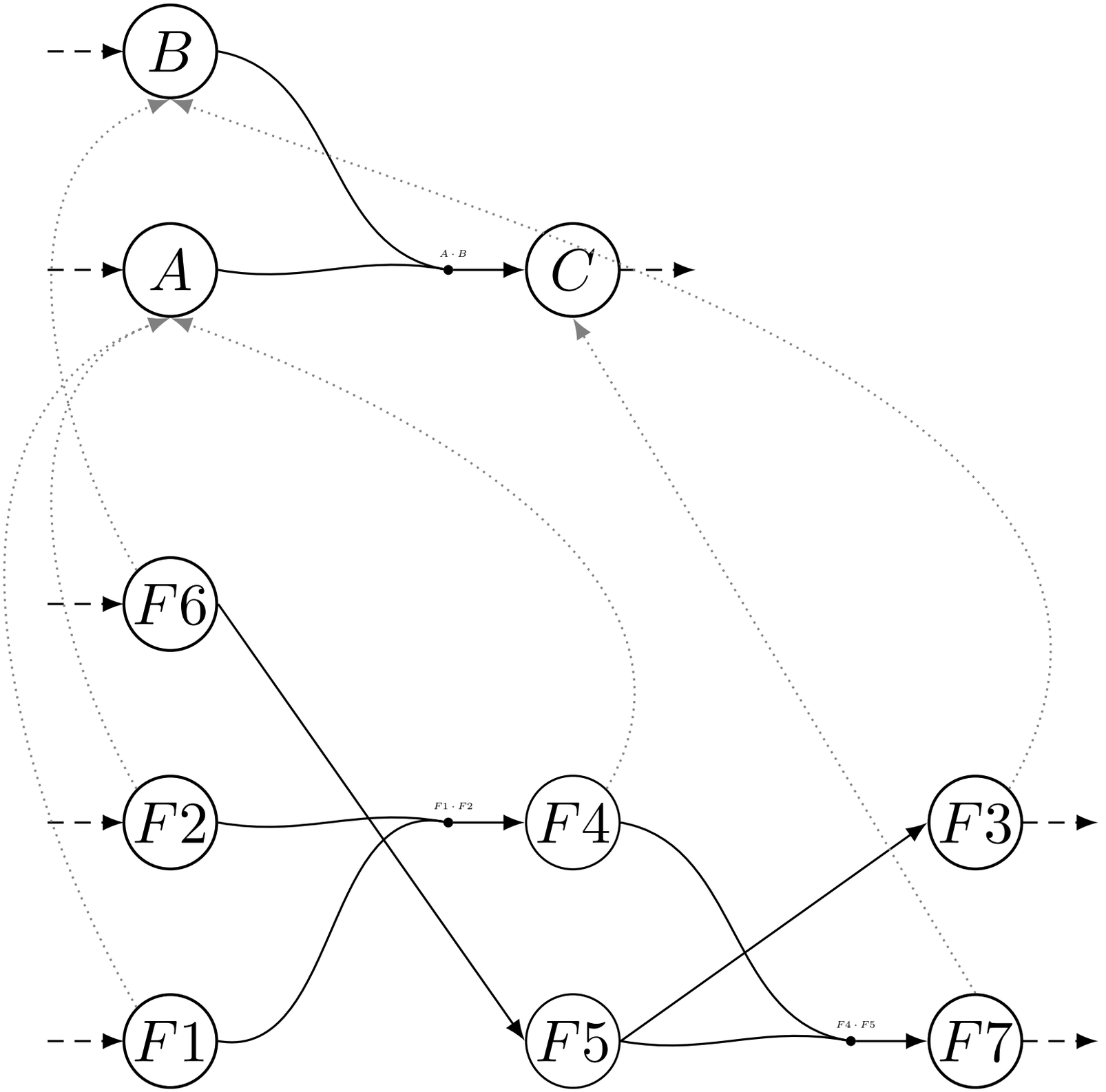
$$(A \cdot B \leftrightarrow C) \cdot (F5 \leftrightarrow F3) \cdot (F1 \cdot F2 \leftrightarrow F4) \cdot (F6 \leftrightarrow F5) \cdot (F3 \cdot F4 \leftrightarrow F7)$$

solution no. 15/54



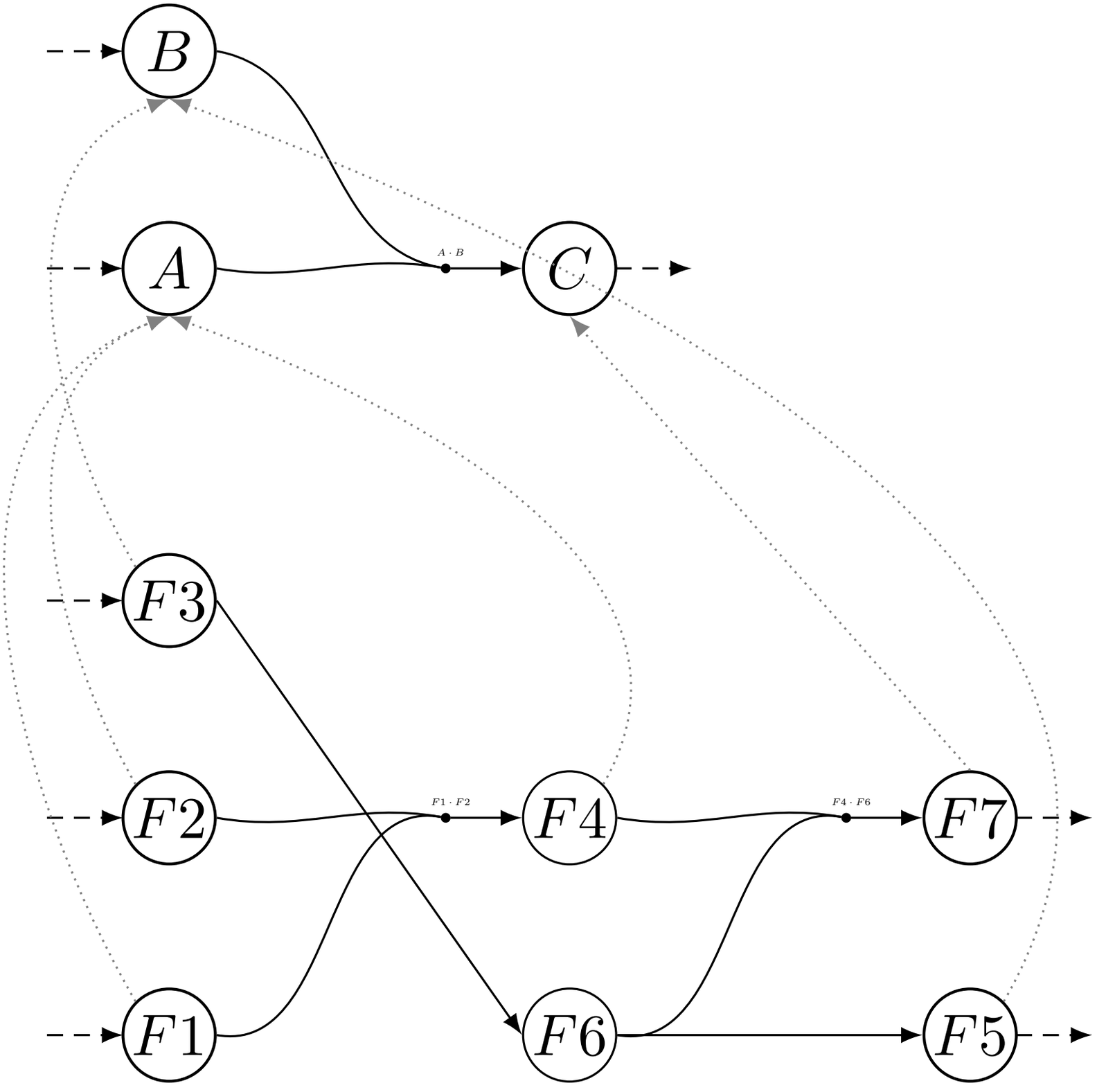
$$(A \cdot B \leftrightarrow C) \cdot (F1 \cdot F2 \leftrightarrow F4) \cdot (F6 \leftrightarrow F5) \cdot (F3 \leftrightarrow F6) \cdot (F4 \cdot F5 \leftrightarrow F7)$$

solution no. 16/54



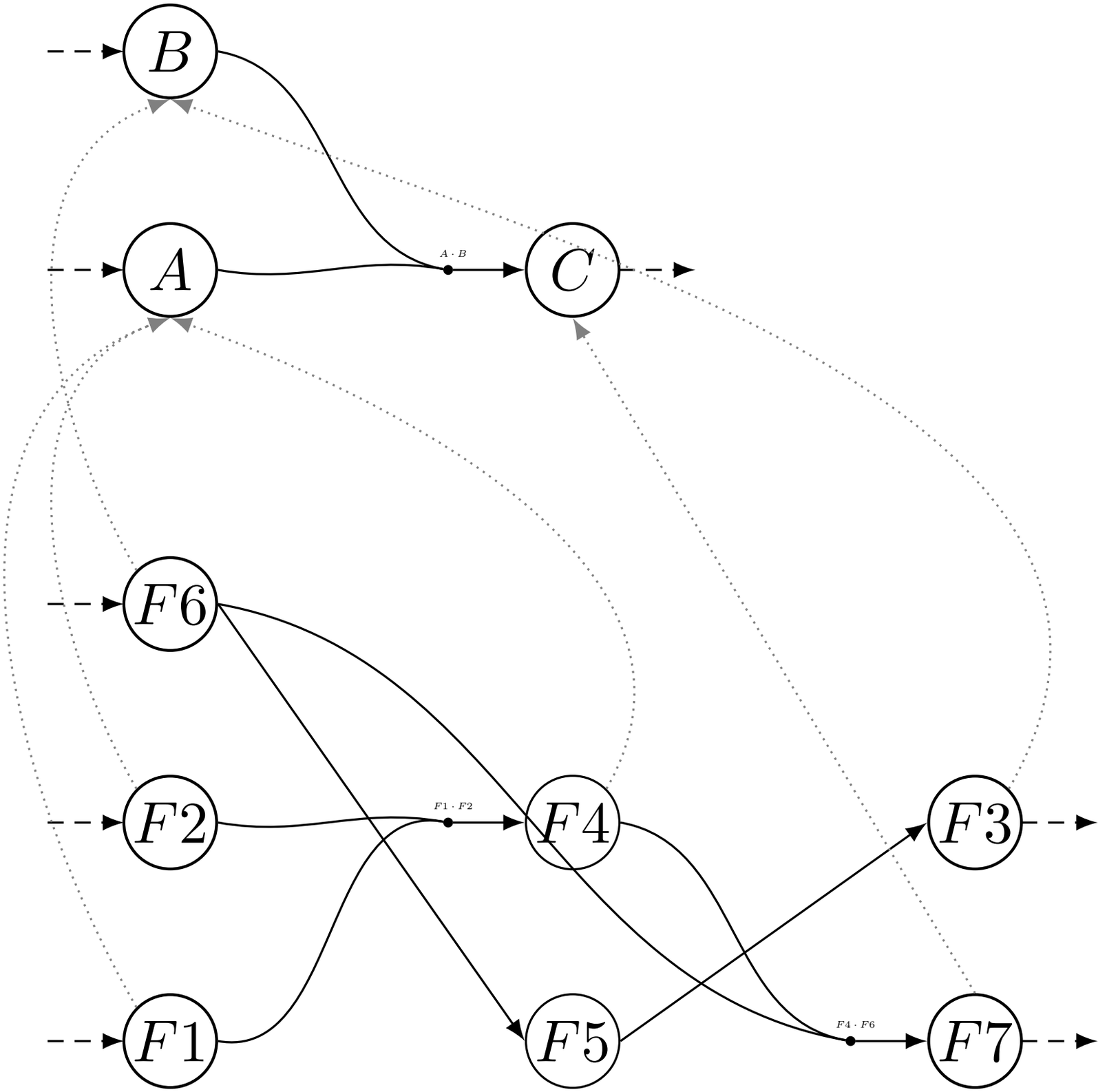
$$(A \cdot B \leftrightarrow C) \cdot (F5 \leftrightarrow F3) \cdot (F1 \cdot F2 \leftrightarrow F4) \cdot (F6 \leftrightarrow F5) \cdot (F4 \cdot F5 \leftrightarrow F7)$$

solution no. 17/54



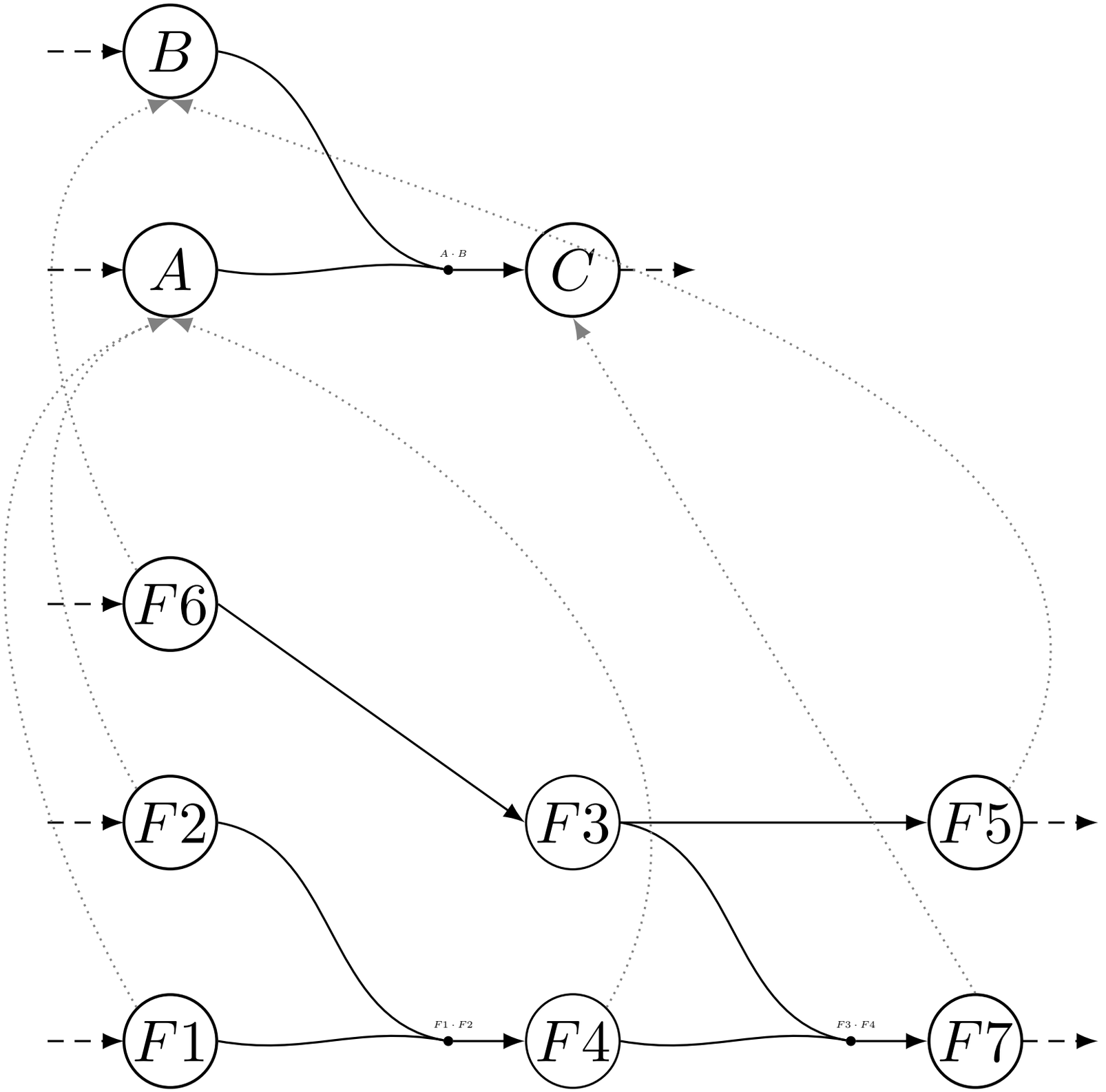
$$(A \cdot B \leftrightarrow C) \cdot (F1 \cdot F2 \leftrightarrow F4) \cdot (F6 \leftrightarrow F5) \cdot (F3 \leftrightarrow F6) \cdot (F4 \cdot F6 \leftrightarrow F7)$$

solution no. 18/54



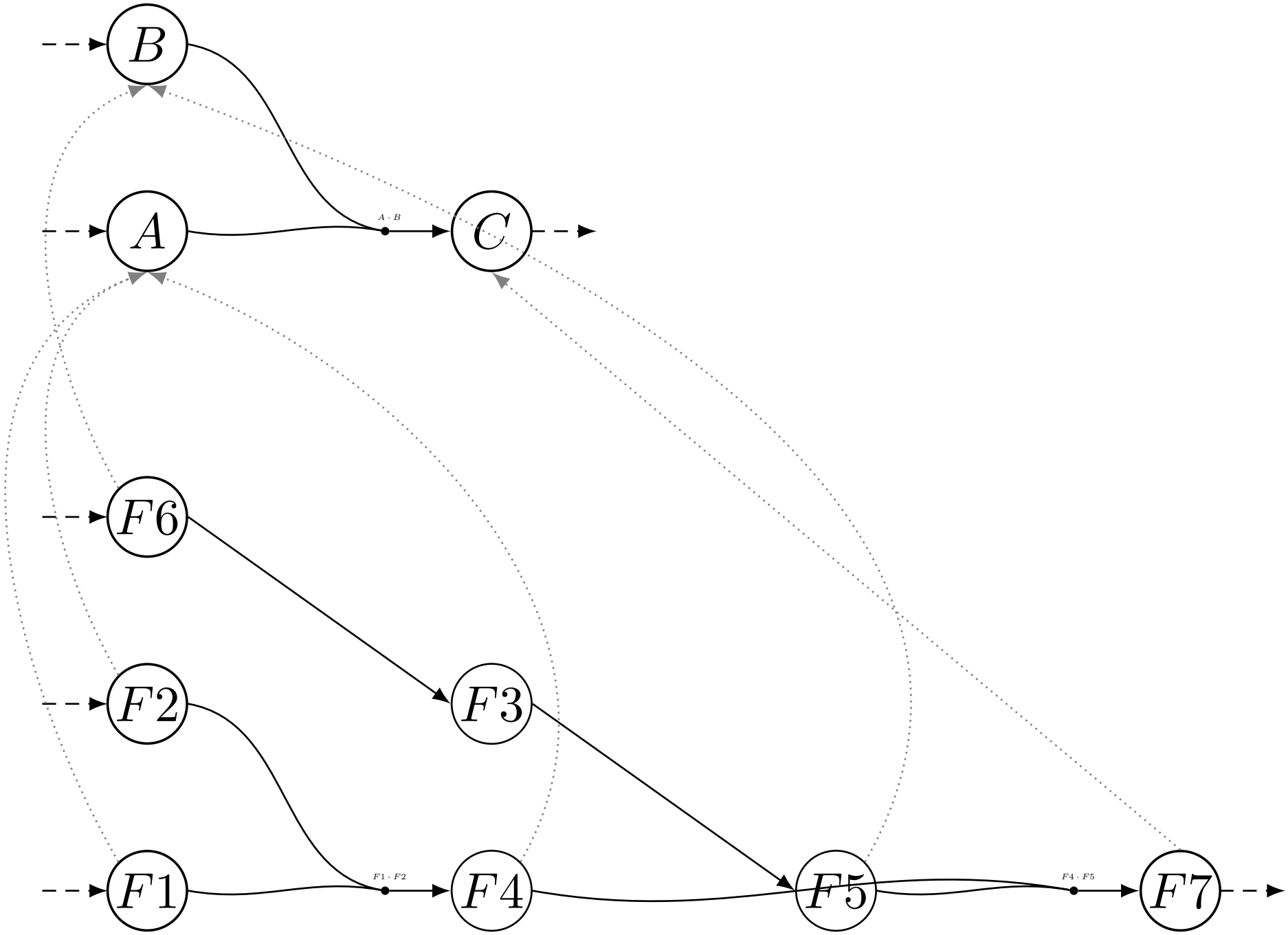
$$(A \cdot B \leftrightarrow C) \cdot (F5 \leftrightarrow F3) \cdot (F1 \cdot F2 \leftrightarrow F4) \cdot (F6 \leftrightarrow F5) \cdot (F4 \cdot F6 \leftrightarrow F7)$$

solution no. 19/54



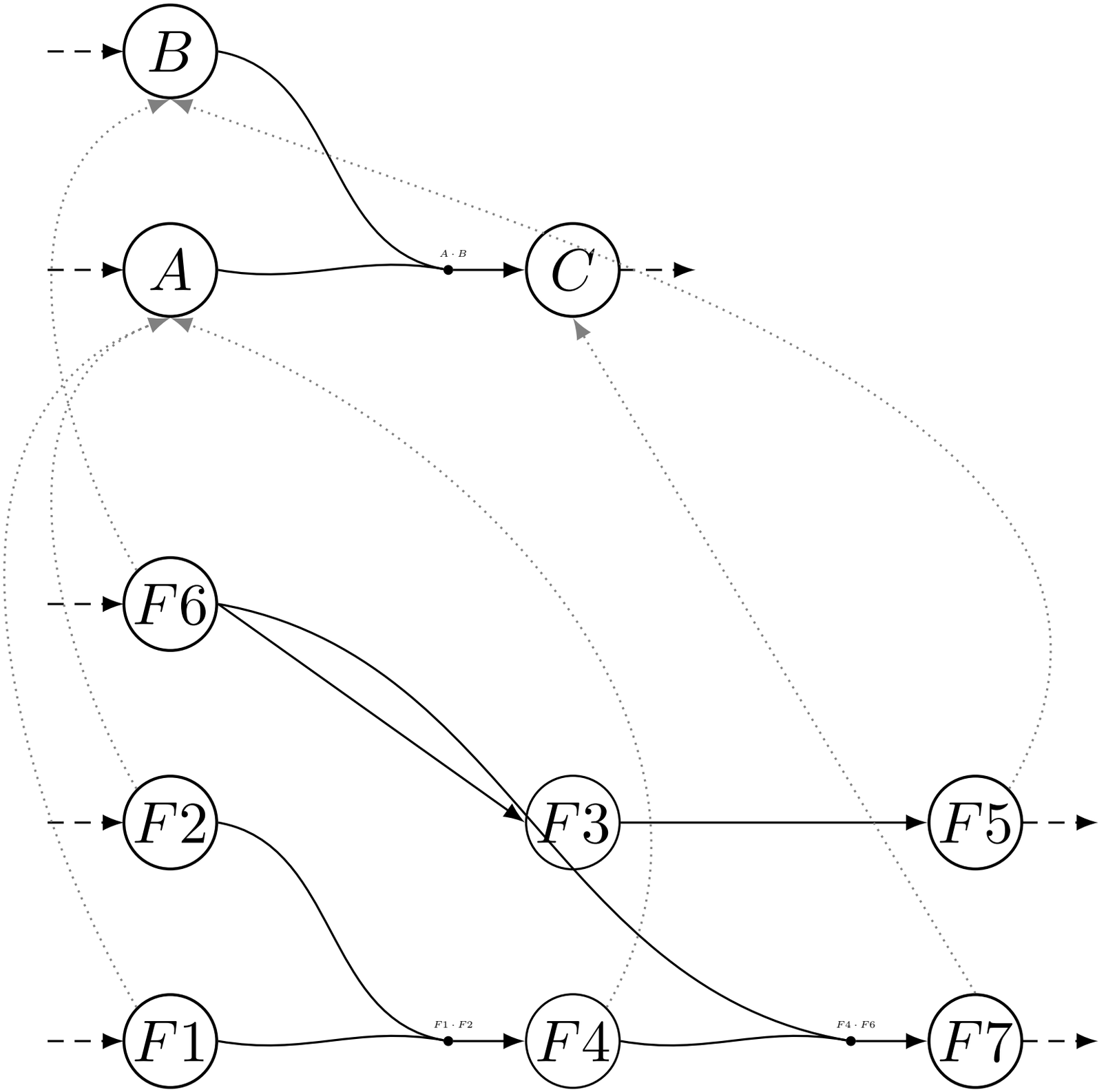
$$(A \cdot B \leftrightarrow C) \cdot (F6 \leftrightarrow F3) \cdot (F1 \cdot F2 \leftrightarrow F4) \cdot (F3 \leftrightarrow F5) \cdot (F3 \cdot F4 \leftrightarrow F7)$$

solution no. 20/54



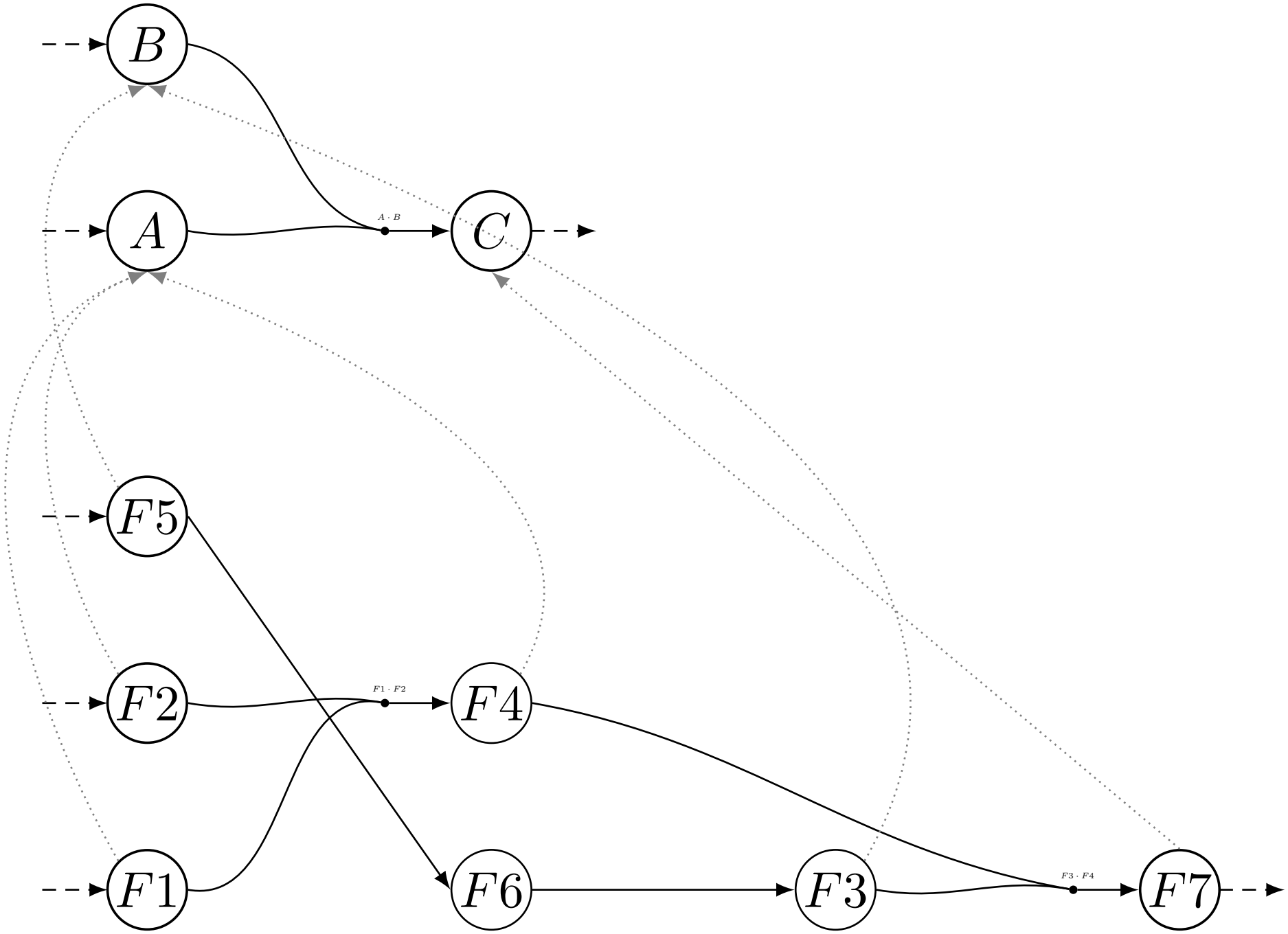
$$(A \cdot B \leftrightarrow C) \cdot (F6 \leftrightarrow F3) \cdot (F1 \cdot F2 \leftrightarrow F4) \cdot (F3 \leftrightarrow F5) \cdot (F4 \cdot F5 \leftrightarrow F7)$$

solution no. 21/54



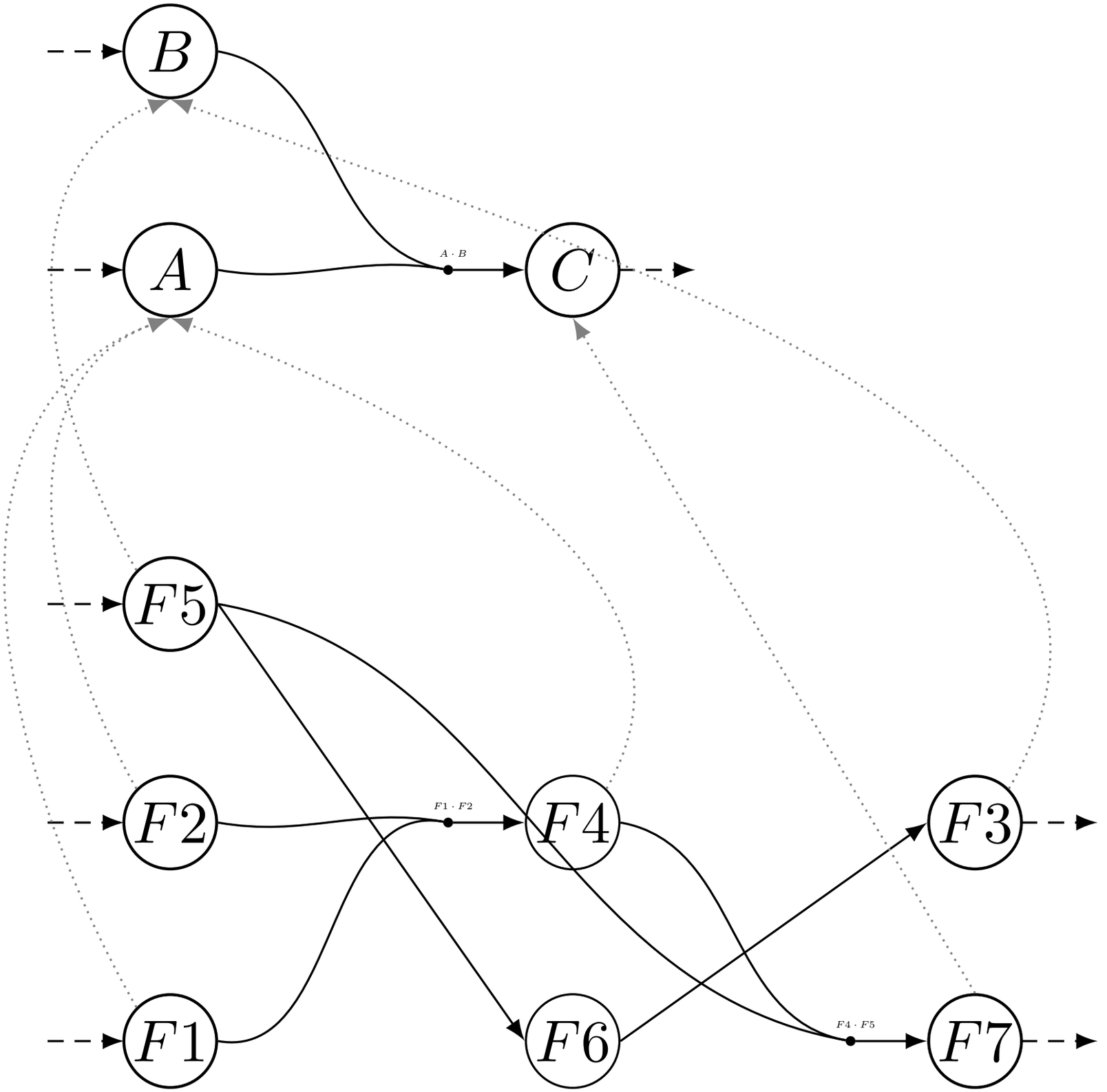
$$(A \cdot B \leftrightarrow C) \cdot (F6 \leftrightarrow F3) \cdot (F1 \cdot F2 \leftrightarrow F4) \cdot (F3 \leftrightarrow F5) \cdot (F4 \cdot F6 \leftrightarrow F7)$$

solution no. 22/54



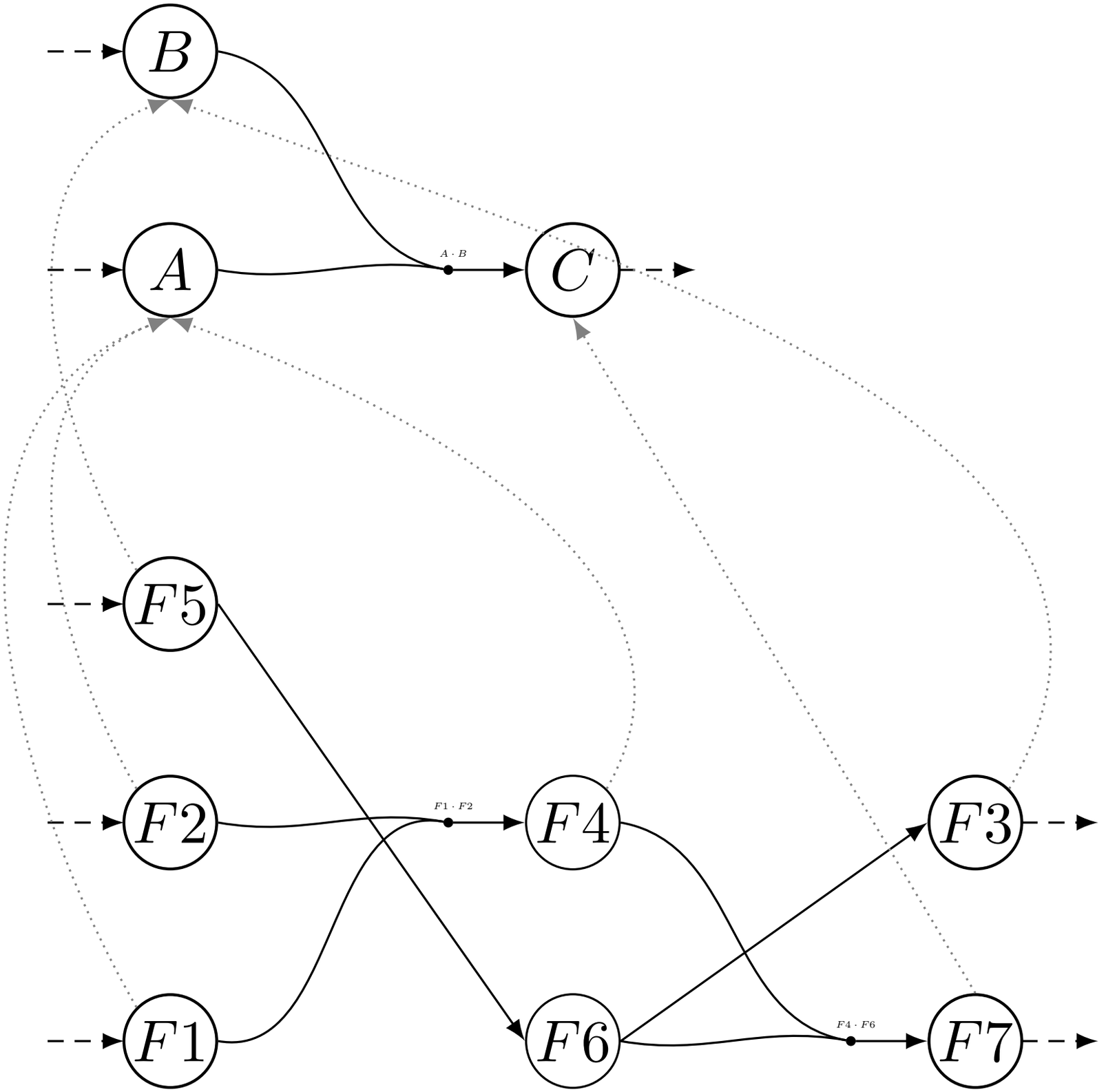
$$(A \cdot B \leftrightarrow C) \cdot (F6 \leftrightarrow F3) \cdot (F1 \cdot F2 \leftrightarrow F4) \cdot (F5 \leftrightarrow F6) \cdot (F3 \cdot F4 \leftrightarrow F7)$$

solution no. 23/54



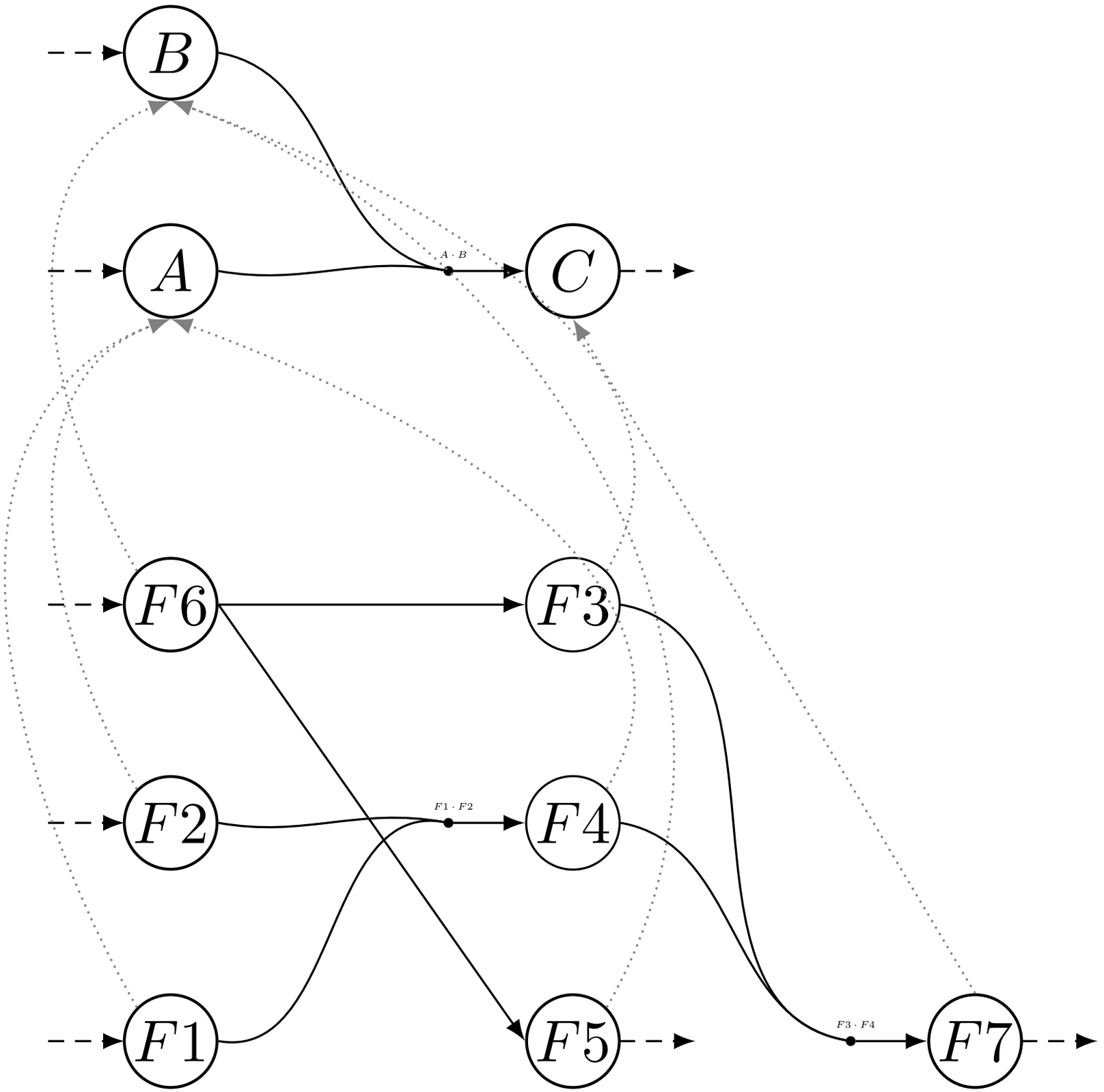
$$(A \cdot B \leftrightarrow C) \cdot (F6 \leftrightarrow F3) \cdot (F1 \cdot F2 \leftrightarrow F4) \cdot (F5 \leftrightarrow F6) \cdot (F4 \cdot F5 \leftrightarrow F7)$$

solution no. 24/54



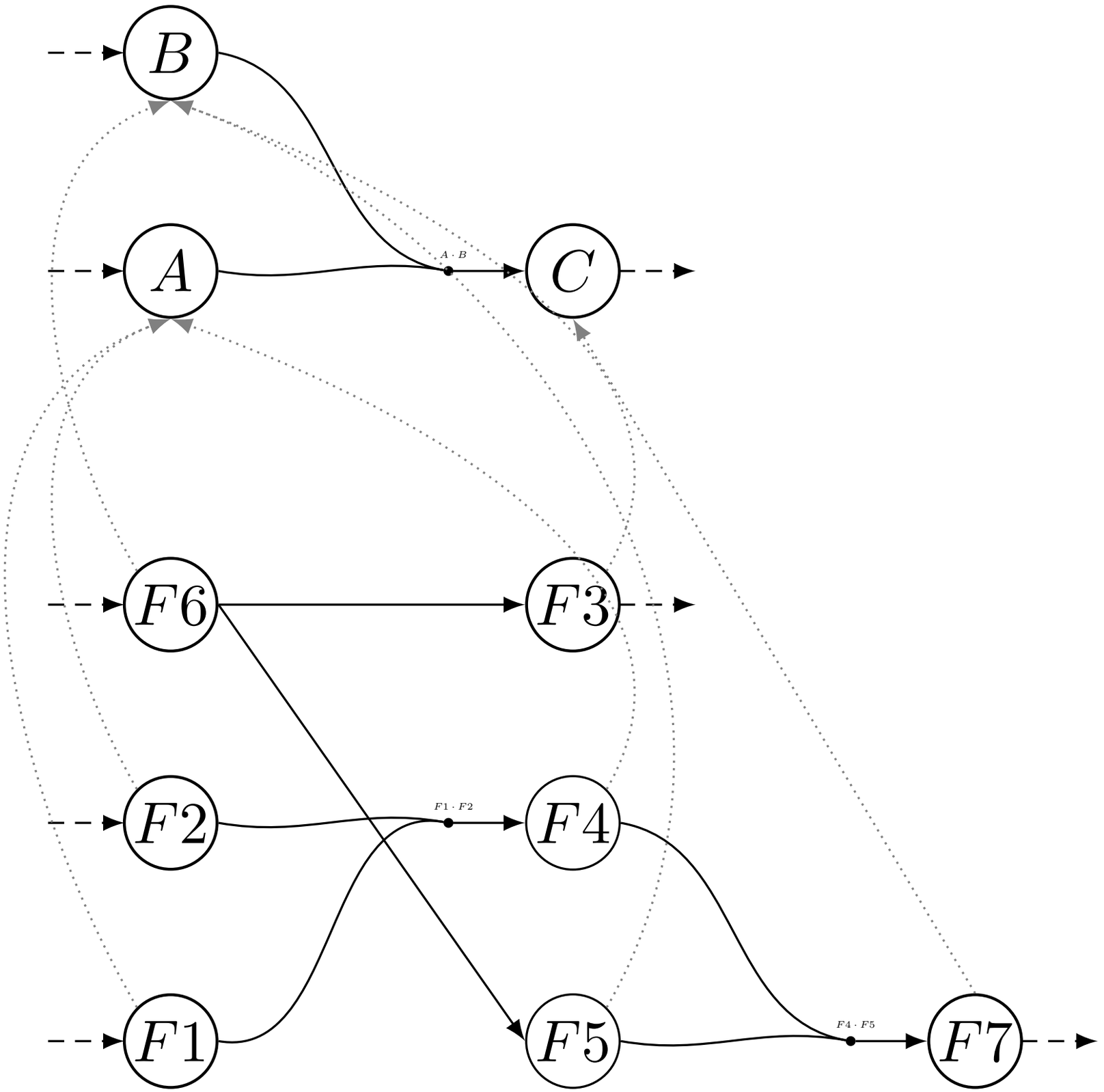
$$(A \cdot B \leftrightarrow C) \cdot (F6 \leftrightarrow F3) \cdot (F1 \cdot F2 \leftrightarrow F4) \cdot (F5 \leftrightarrow F6) \cdot (F4 \cdot F6 \leftrightarrow F7)$$

solution no. 25/54



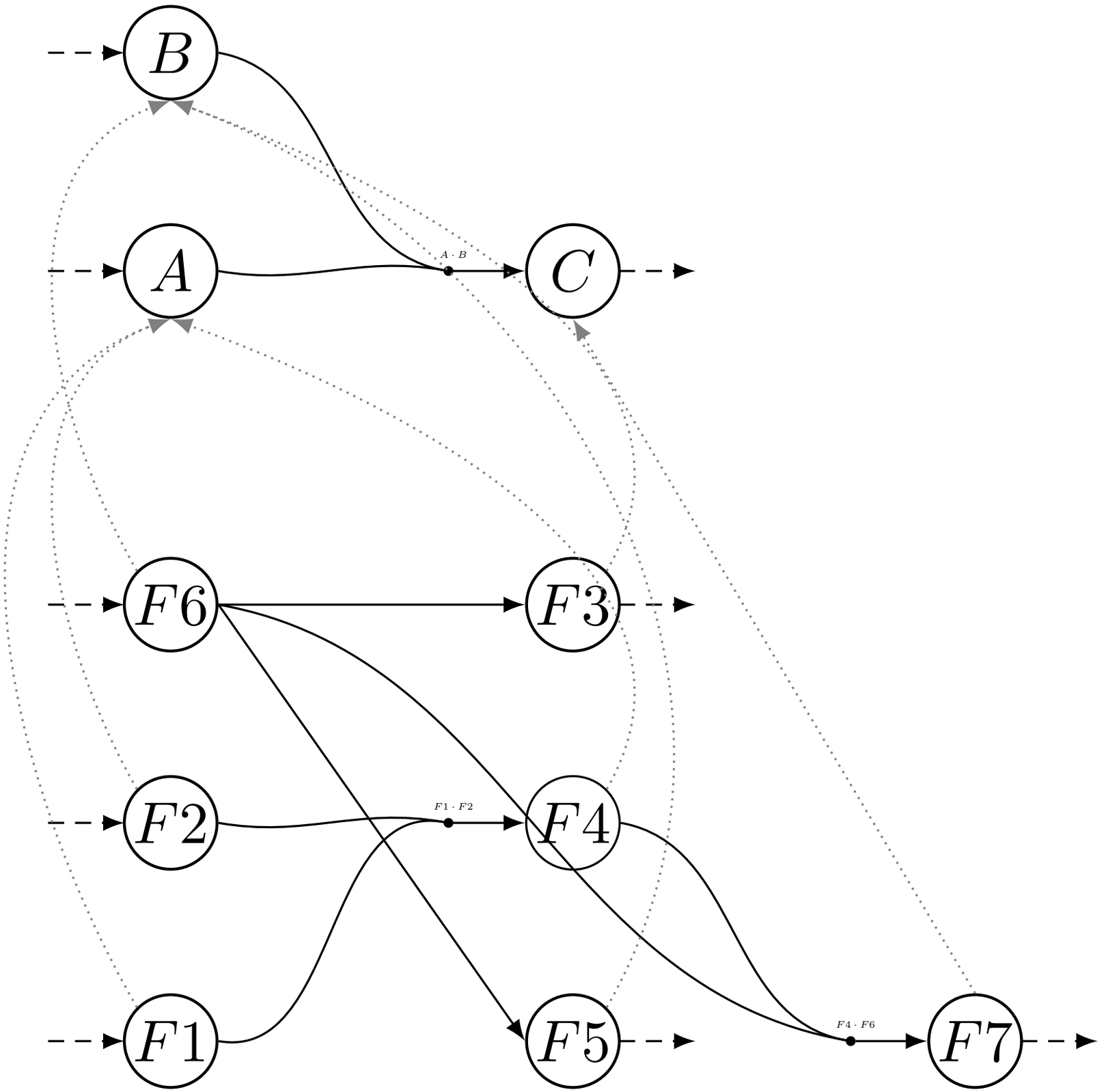
$$(A \cdot B \leftrightarrow C) \cdot (F6 \leftrightarrow F3) \cdot (F1 \cdot F2 \leftrightarrow F4) \cdot (F6 \leftrightarrow F5) \cdot (F3 \cdot F4 \leftrightarrow F7)$$

solution no. 26/54



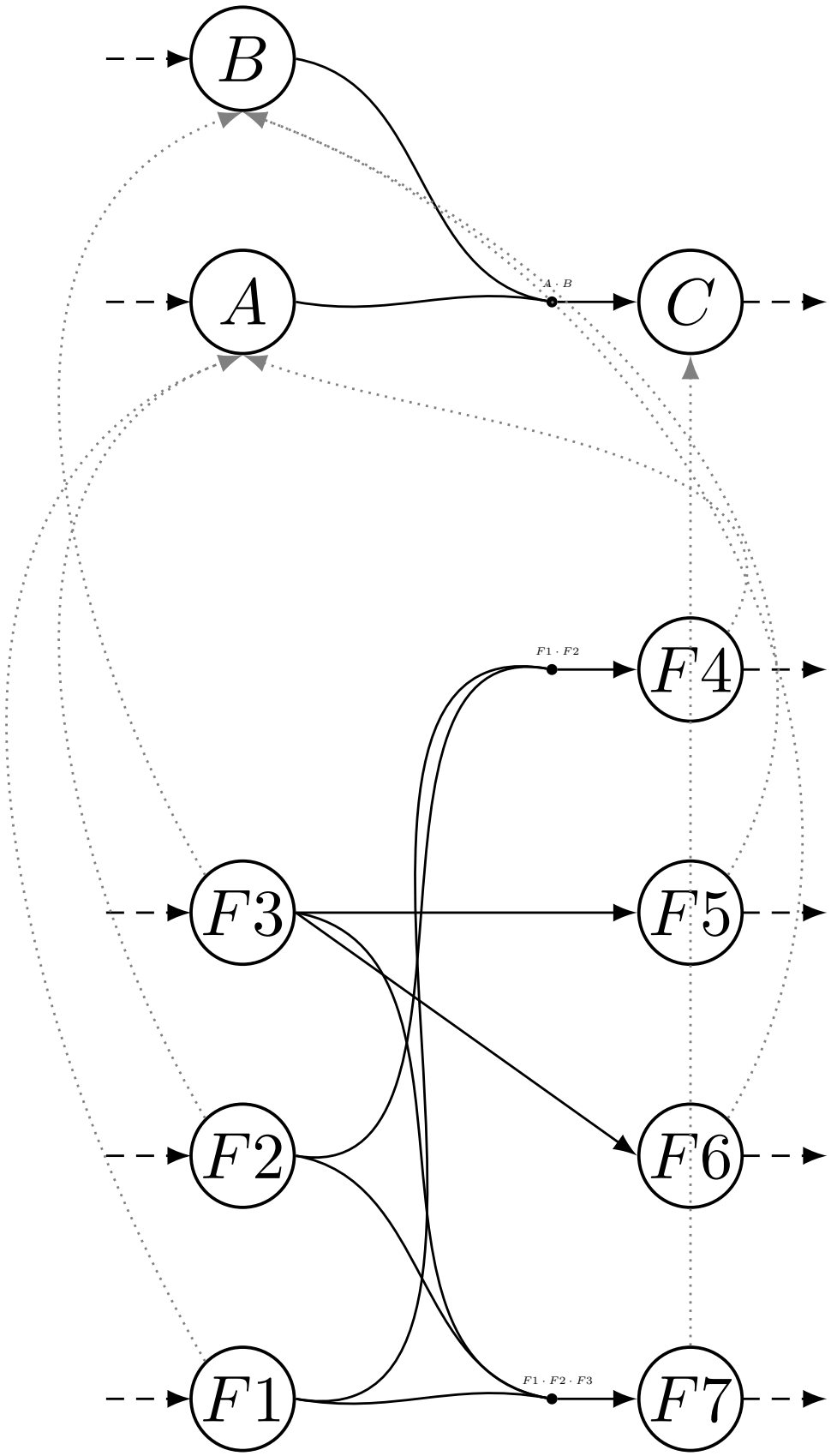
$$(A \cdot B \leftrightarrow C) \cdot (F6 \leftrightarrow F3) \cdot (F1 \cdot F2 \leftrightarrow F4) \cdot (F6 \leftrightarrow F5) \cdot (F4 \cdot F5 \leftrightarrow F7)$$

solution no. 27/54



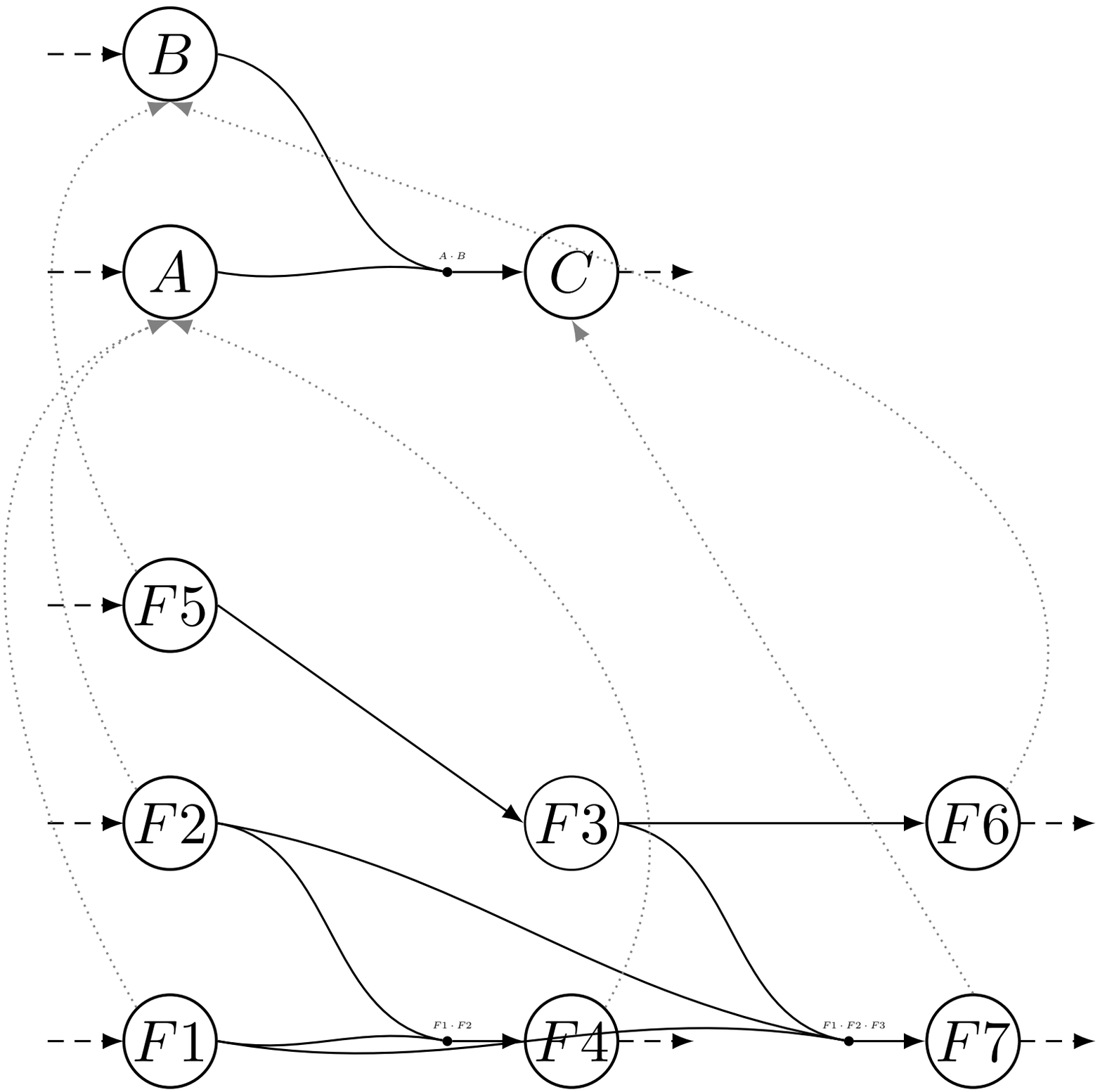
$$(A \cdot B \leftrightarrow C) \cdot (F6 \leftrightarrow F3) \cdot (F1 \cdot F2 \leftrightarrow F4) \cdot (F6 \leftrightarrow F5) \cdot (F4 \cdot F6 \leftrightarrow F7)$$

solution no. 28/54



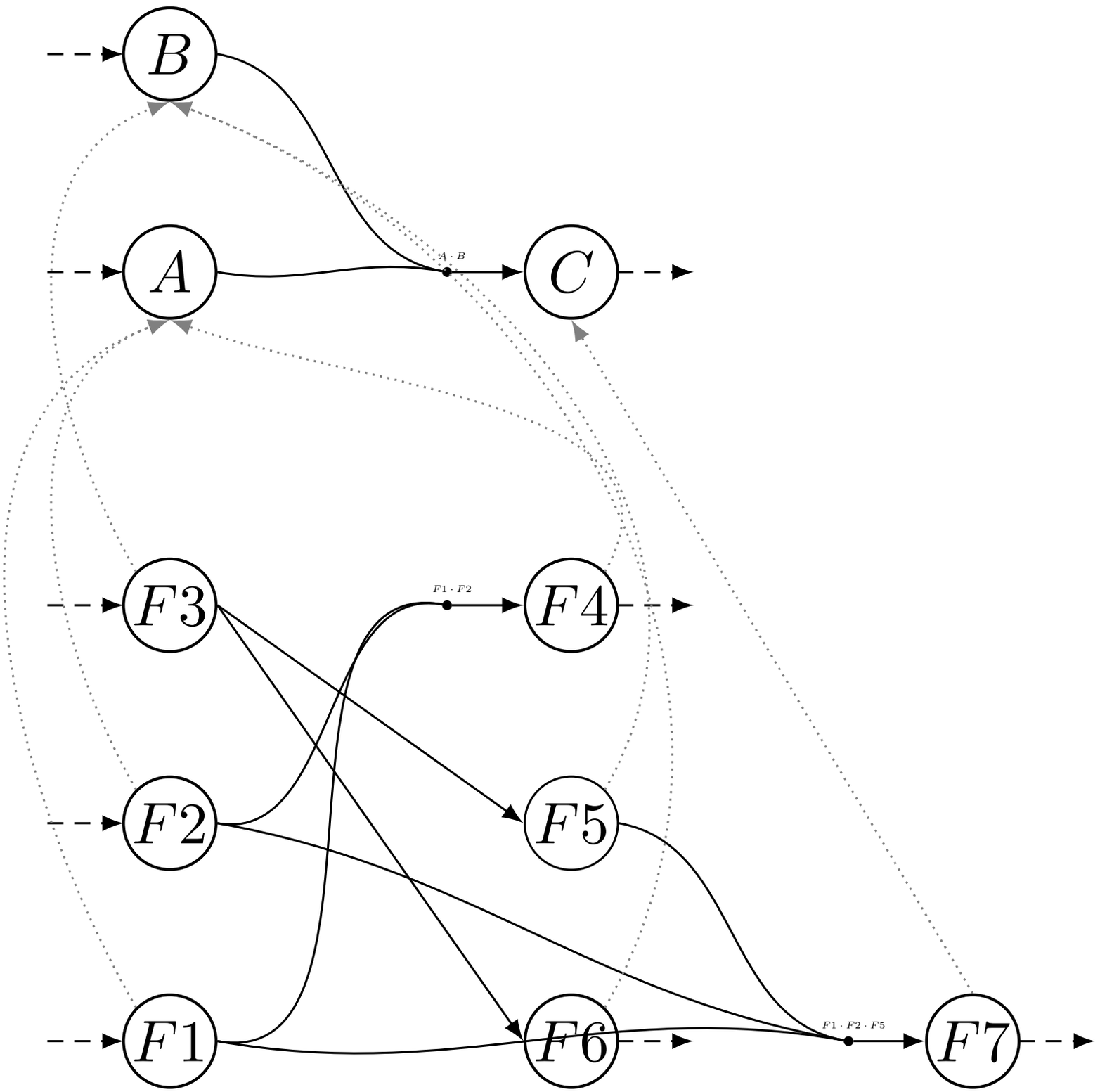
$$(A \cdot B \leftrightarrow C) \cdot (F1 \cdot F2 \leftrightarrow F4) \cdot (F3 \leftrightarrow F5) \cdot (F3 \leftrightarrow F6) \cdot (F1 \cdot F2 \cdot F3 \leftrightarrow F7)$$

solution no. 29/54



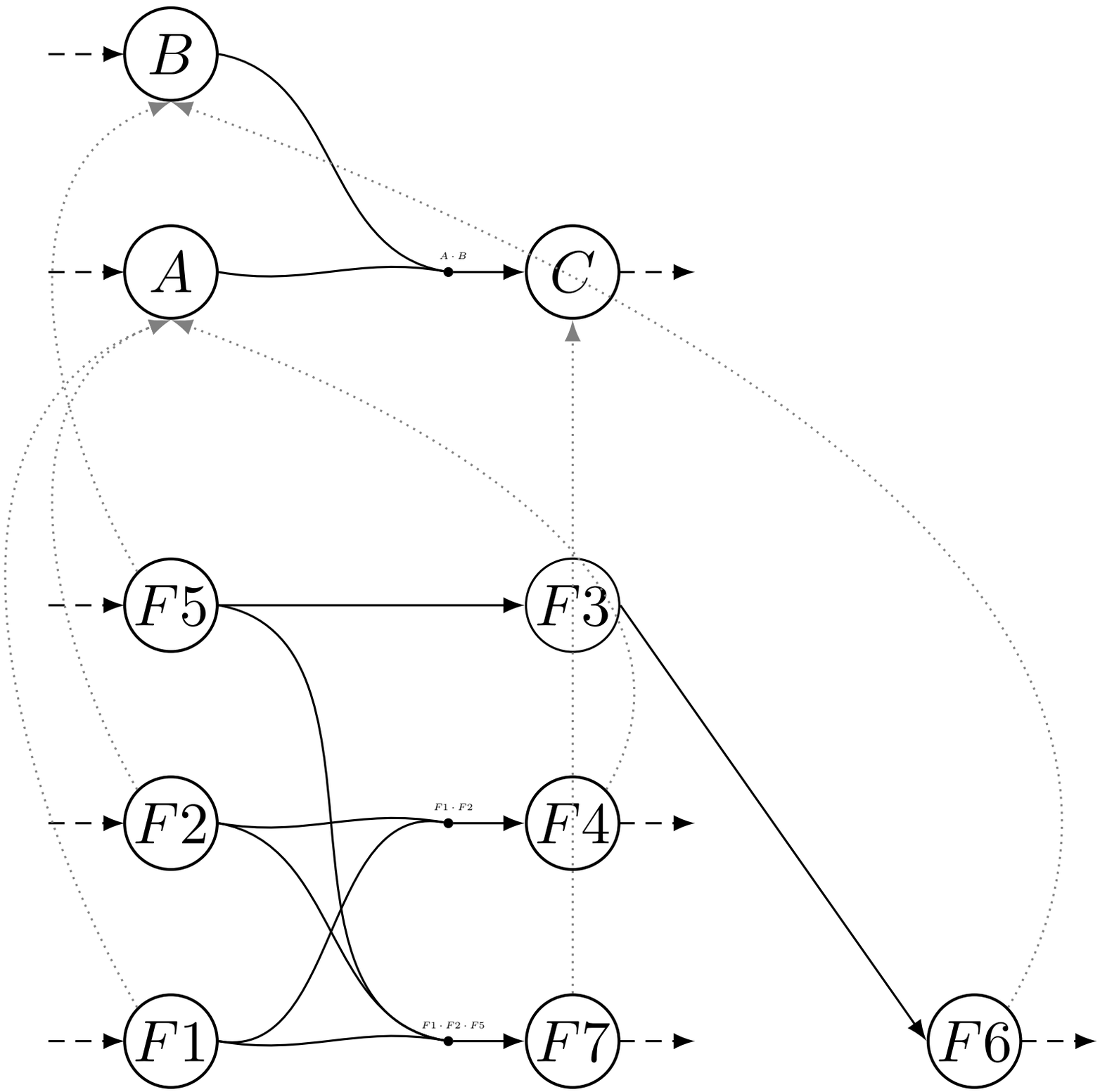
$$(A \cdot B \leftrightarrow C) \cdot (F5 \leftrightarrow F3) \cdot (F1 \cdot F2 \leftrightarrow F4) \cdot (F3 \leftrightarrow F6) \cdot (F1 \cdot F2 \cdot F3 \leftrightarrow F7)$$

solution no. 30/54



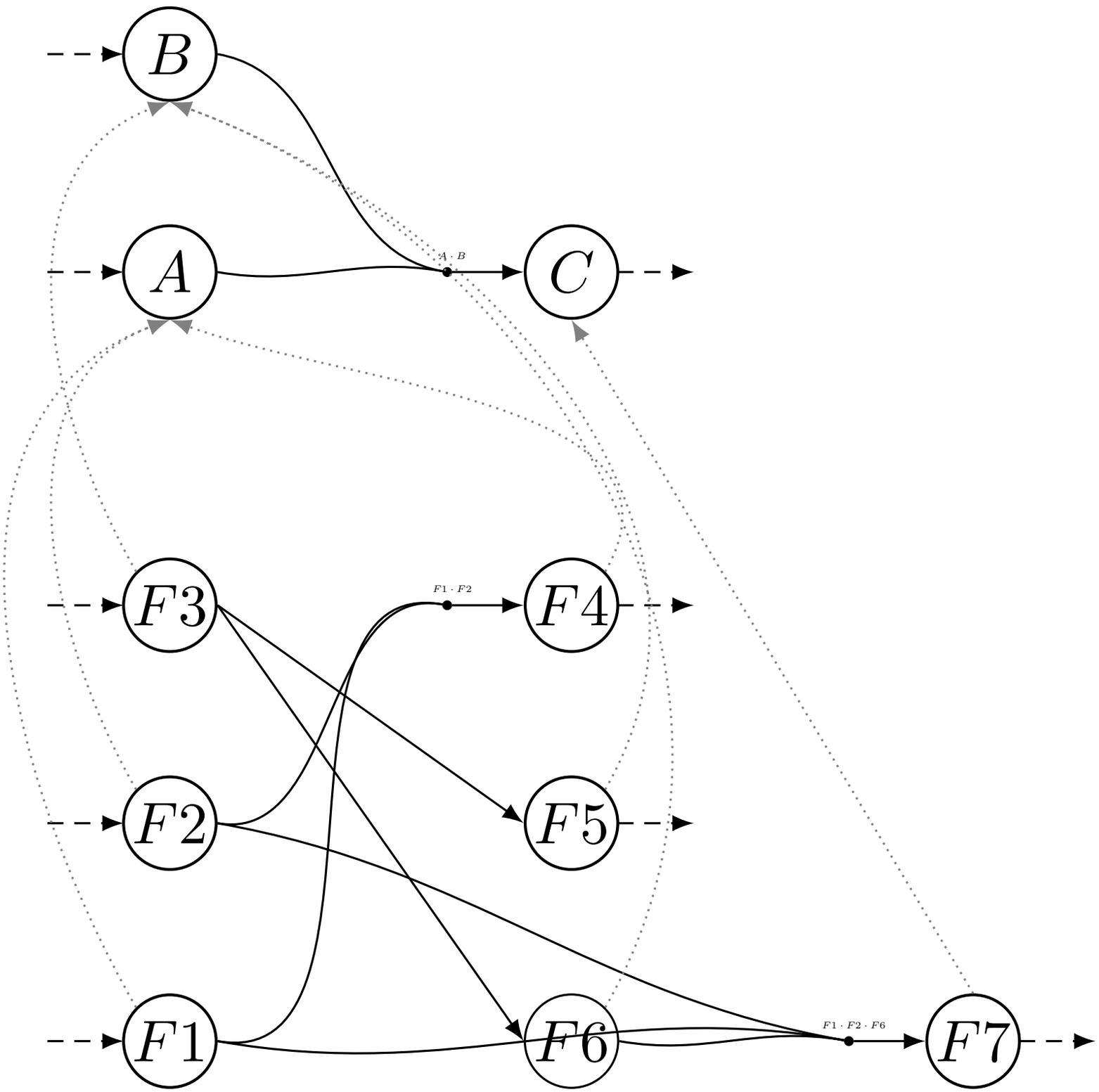
$$(A \cdot B \leftrightarrow C) \cdot (F1 \cdot F2 \leftrightarrow F4) \cdot (F3 \leftrightarrow F5) \cdot (F3 \leftrightarrow F6) \cdot (F1 \cdot F2 \cdot F5 \leftrightarrow F7)$$

solution no. 31/54



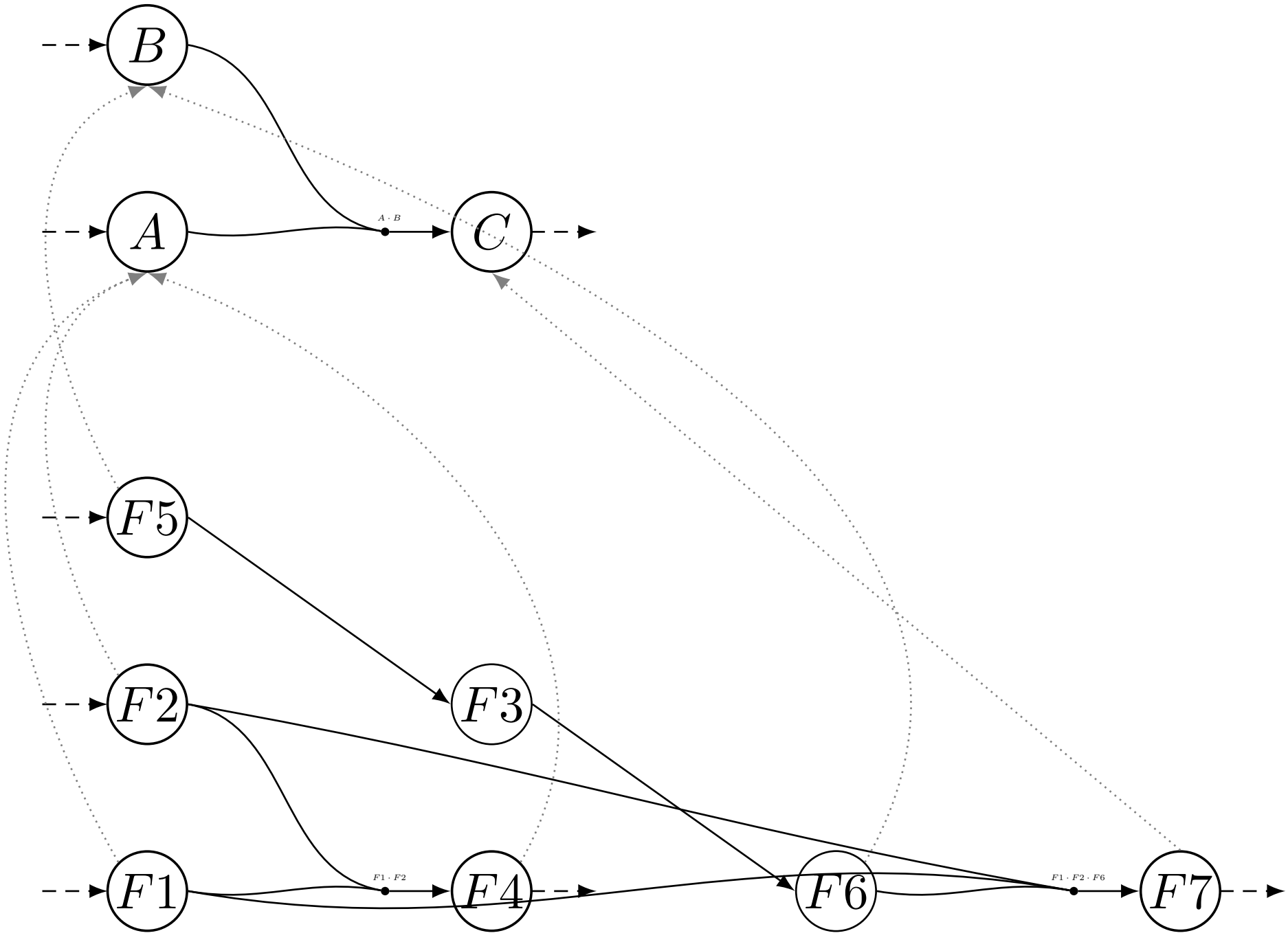
$$(A \cdot B \leftrightarrow C) \cdot (F5 \leftrightarrow F3) \cdot (F1 \cdot F2 \leftrightarrow F4) \cdot (F3 \leftrightarrow F6) \cdot (F1 \cdot F2 \cdot F5 \leftrightarrow F7)$$

solution no. 32/54



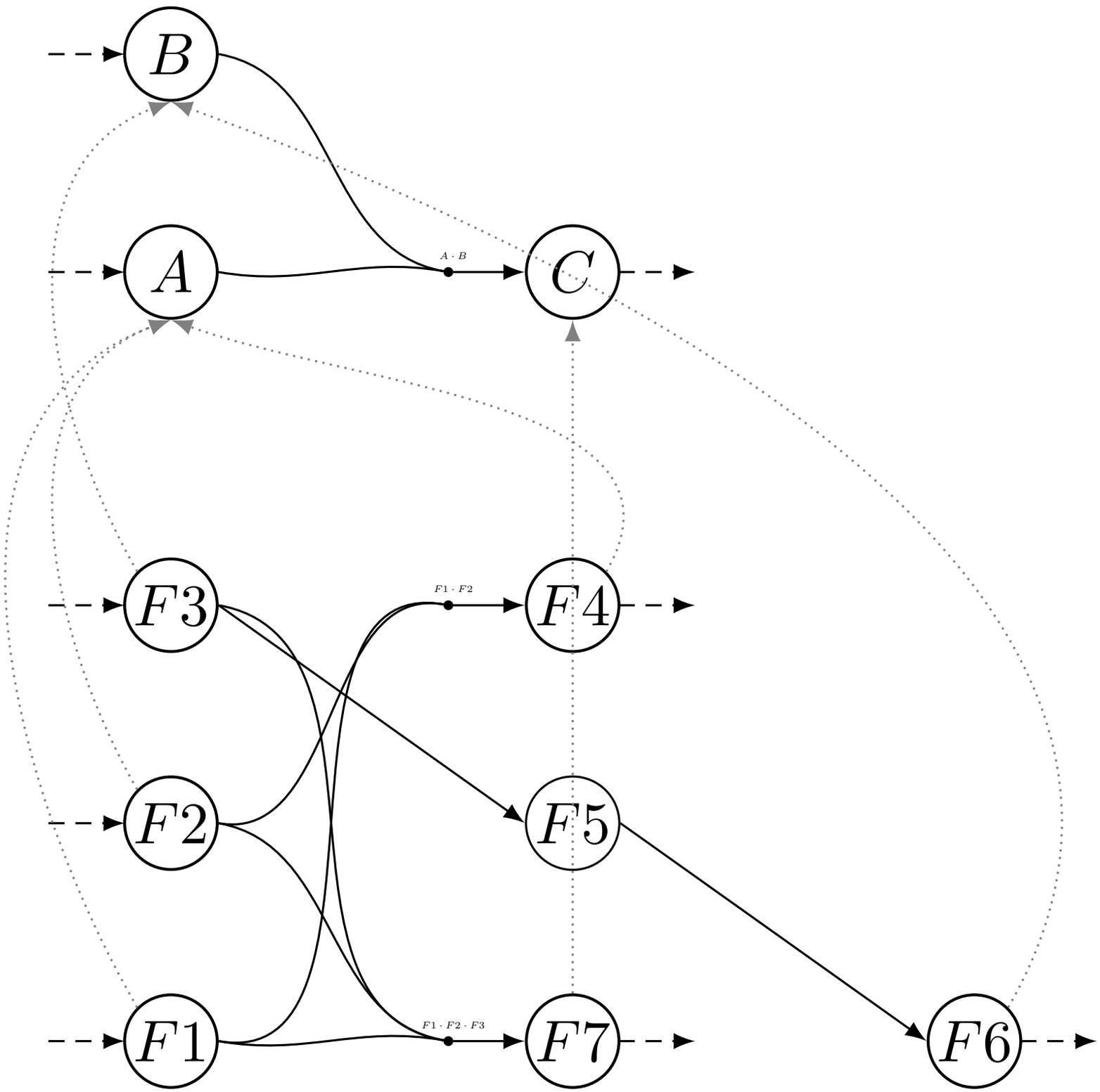
$$(A \cdot B \leftrightarrow C) \cdot (F1 \cdot F2 \leftrightarrow F4) \cdot (F3 \leftrightarrow F5) \cdot (F3 \leftrightarrow F6) \cdot (F1 \cdot F2 \cdot F6 \leftrightarrow F7)$$

solution no. 33/54



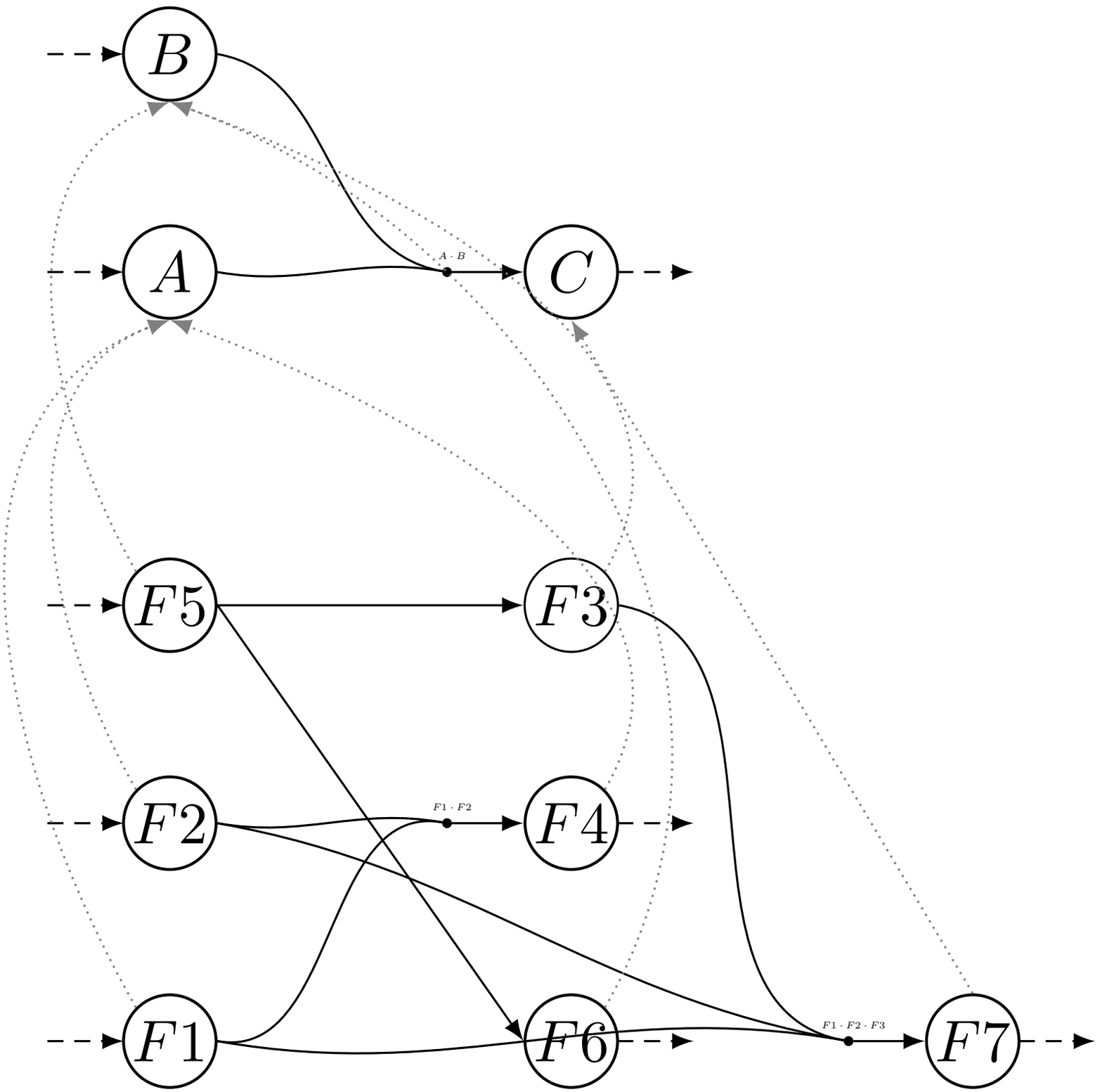
$$(A \cdot B \leftrightarrow C) \cdot (F5 \leftrightarrow F3) \cdot (F1 \cdot F2 \leftrightarrow F4) \cdot (F3 \leftrightarrow F6) \cdot (F1 \cdot F2 \cdot F6 \leftrightarrow F7)$$

solution no. 34/54



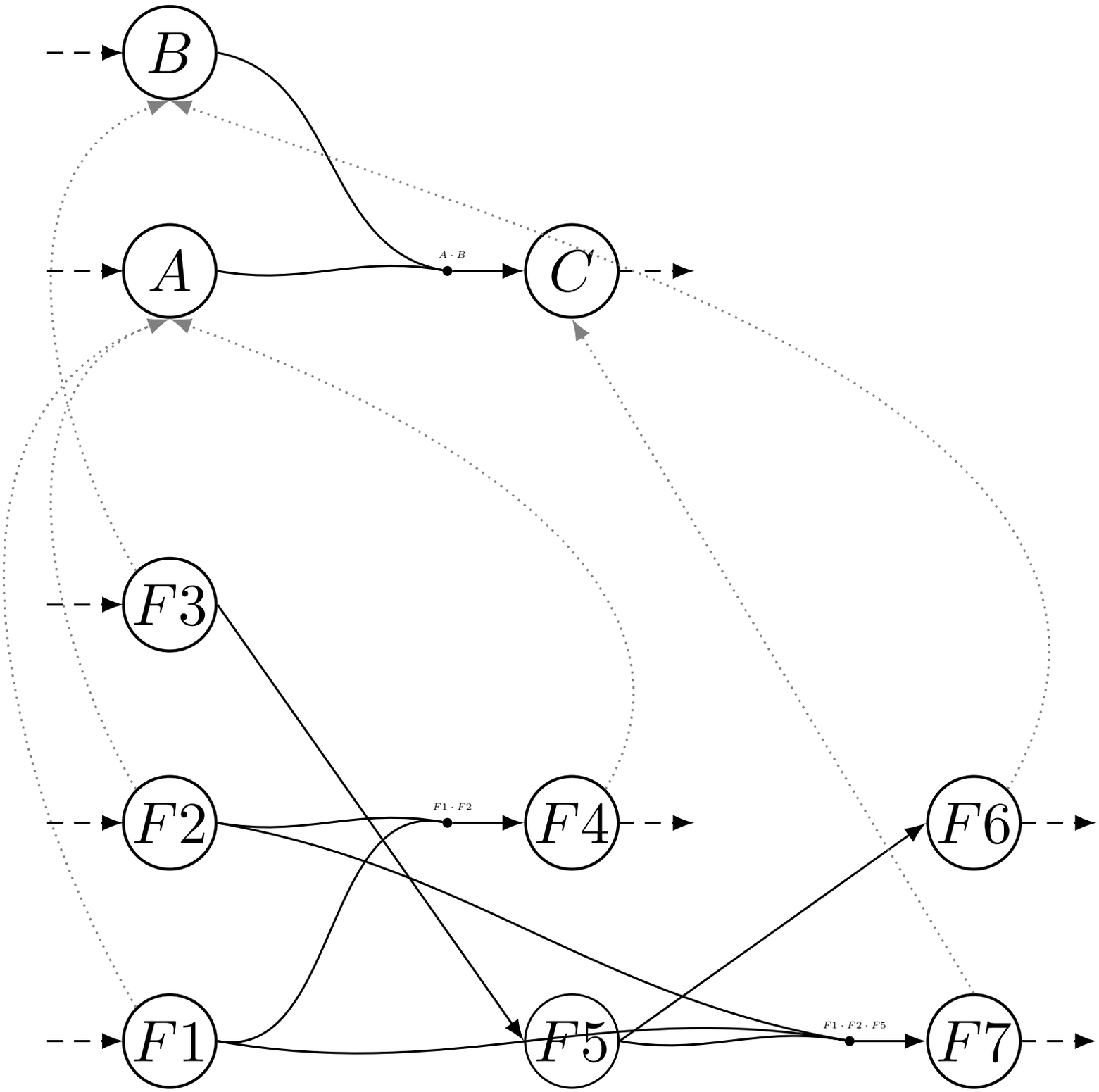
$$(A \cdot B \leftrightarrow C) \cdot (F1 \cdot F2 \leftrightarrow F4) \cdot (F3 \leftrightarrow F5) \cdot (F5 \leftrightarrow F6) \cdot (F1 \cdot F2 \cdot F3 \leftrightarrow F7)$$

solution no. 35/54



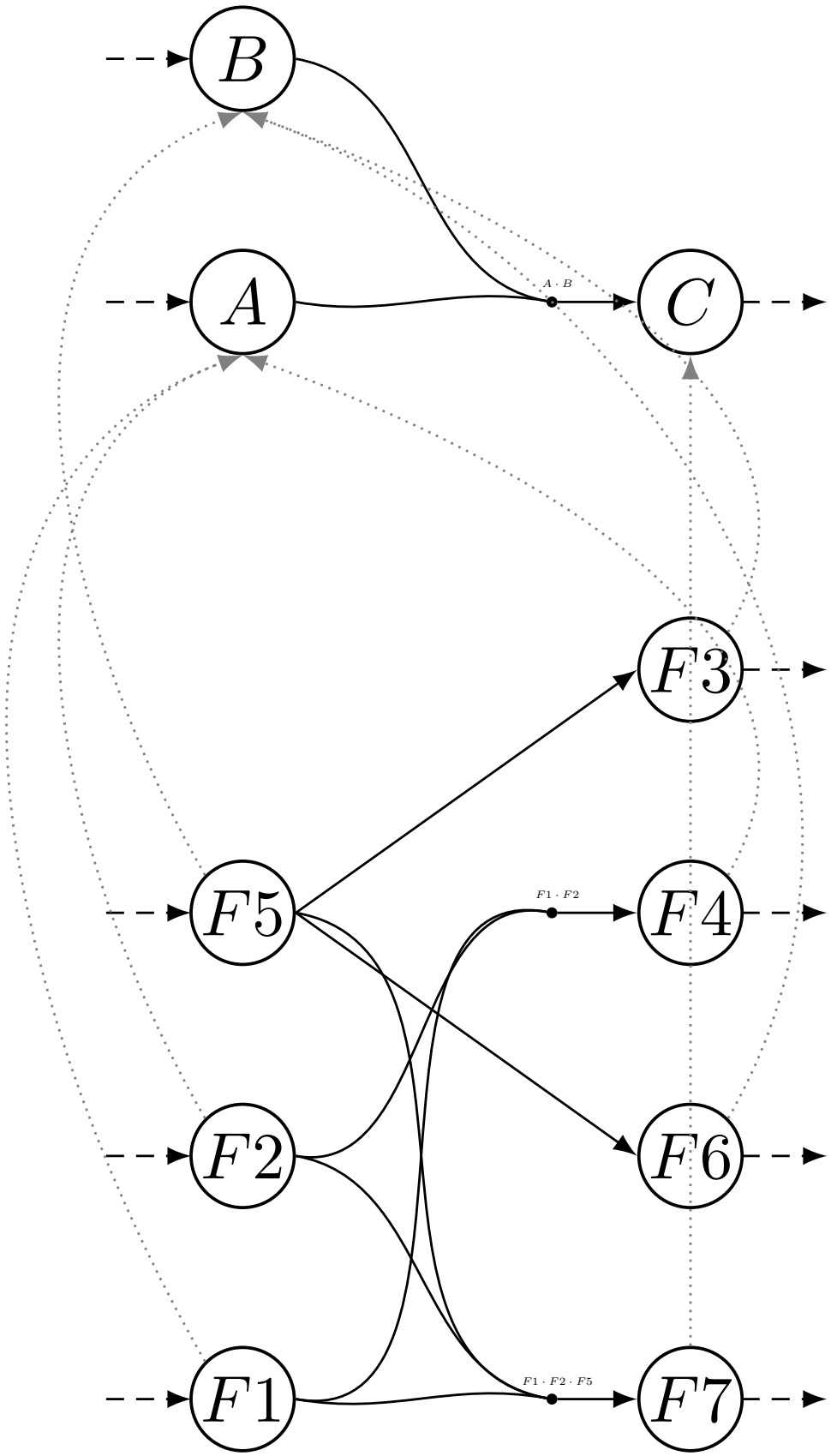
$$(A \cdot B \leftrightarrow C) \cdot (F5 \leftrightarrow F3) \cdot (F1 \cdot F2 \leftrightarrow F4) \cdot (F5 \leftrightarrow F6) \cdot (F1 \cdot F2 \cdot F3 \leftrightarrow F7)$$

solution no. 36/54



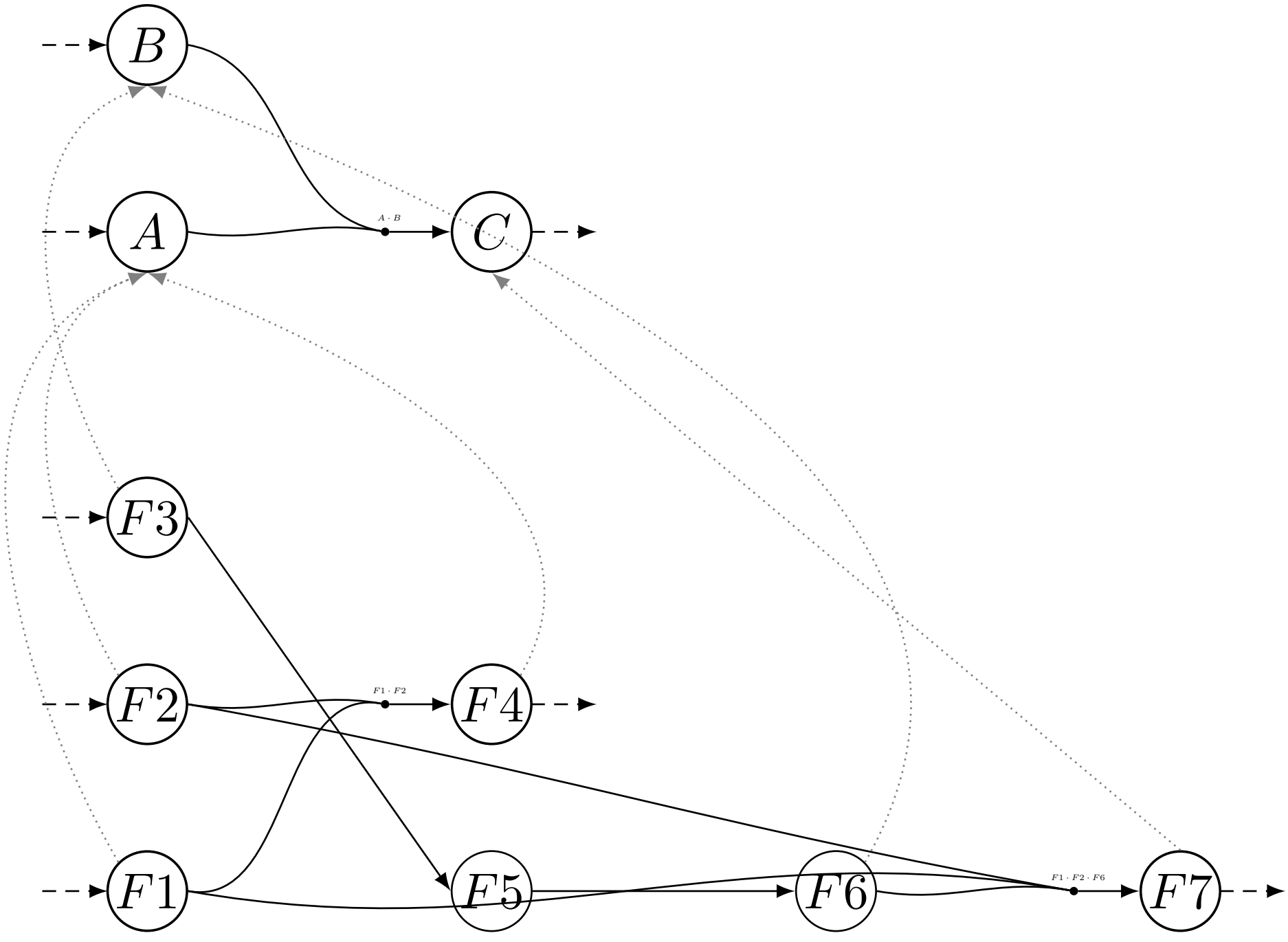
$$(A \cdot B \leftrightarrow C) \cdot (F1 \cdot F2 \leftrightarrow F4) \cdot (F3 \leftrightarrow F5) \cdot (F5 \leftrightarrow F6) \cdot (F1 \cdot F2 \cdot F5 \leftrightarrow F7)$$

solution no. 37/54



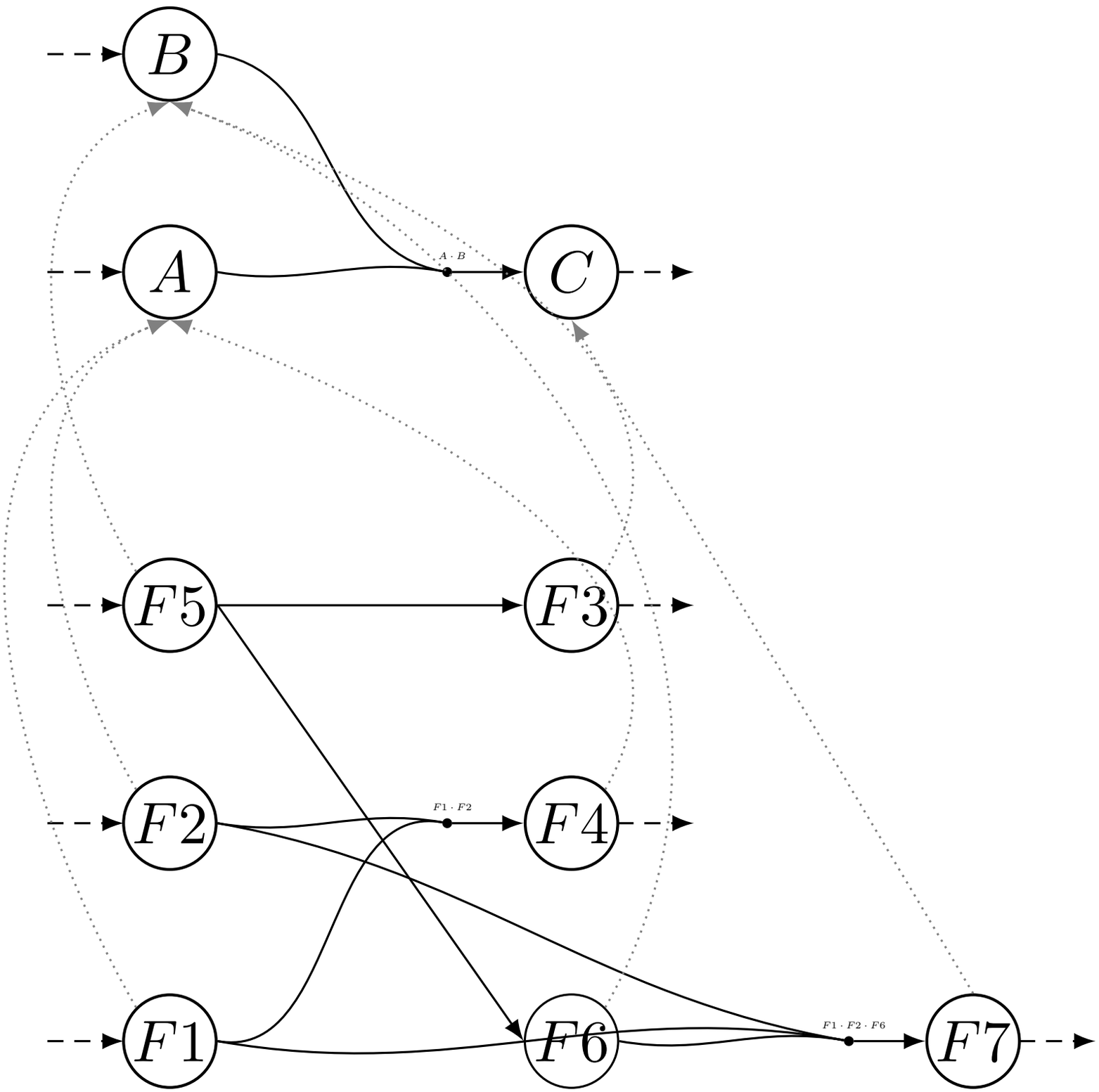
$$(A \cdot B \leftrightarrow C) \cdot (F5 \leftrightarrow F3) \cdot (F1 \cdot F2 \leftrightarrow F4) \cdot (F5 \leftrightarrow F6) \cdot (F1 \cdot F2 \cdot F5 \leftrightarrow F7)$$

solution no. 38/54



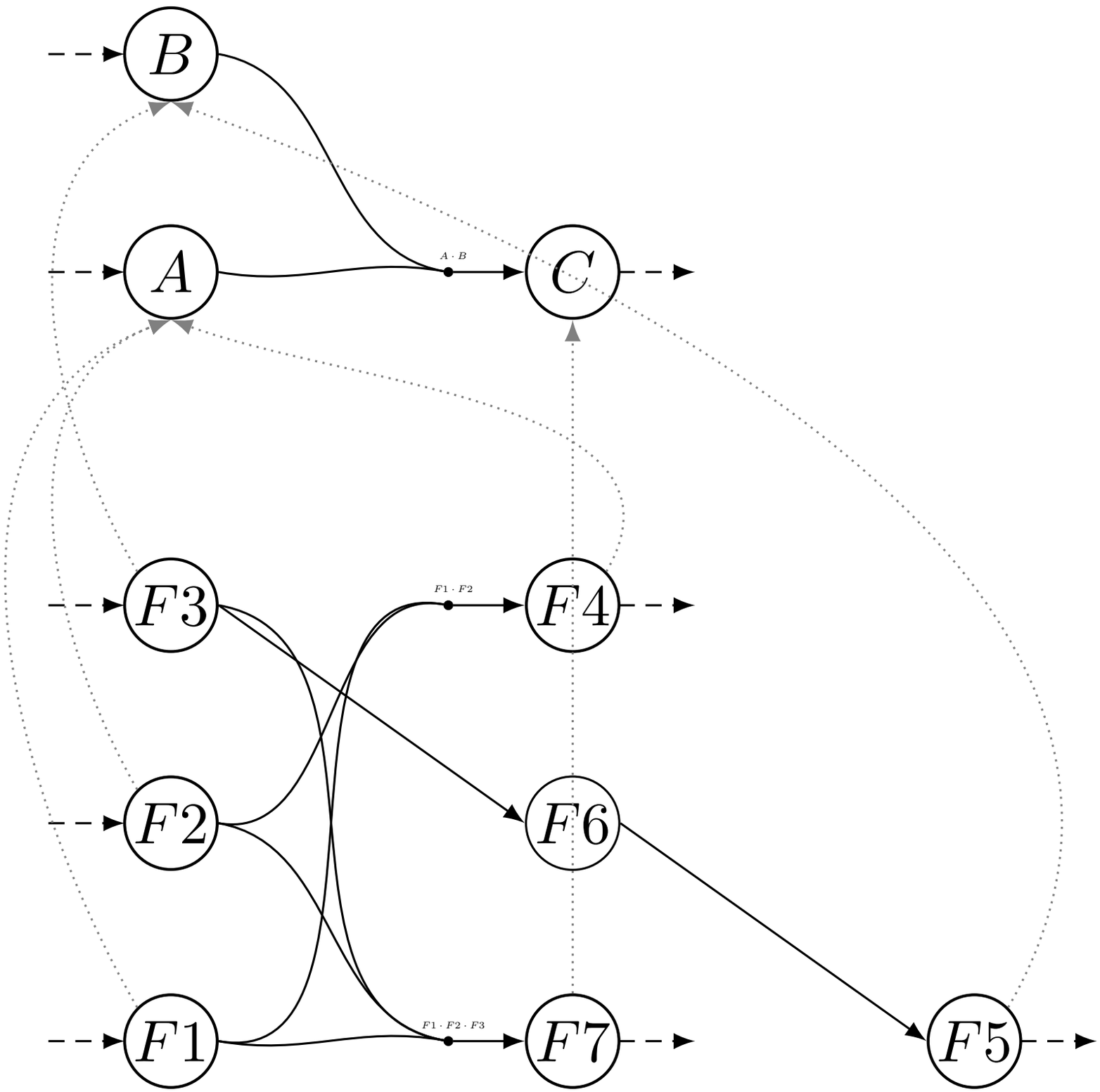
$$(A \cdot B \leftrightarrow C) \cdot (F1 \cdot F2 \leftrightarrow F4) \cdot (F3 \leftrightarrow F5) \cdot (F5 \leftrightarrow F6) \cdot (F1 \cdot F2 \cdot F6 \leftrightarrow F7)$$

solution no. 39/54



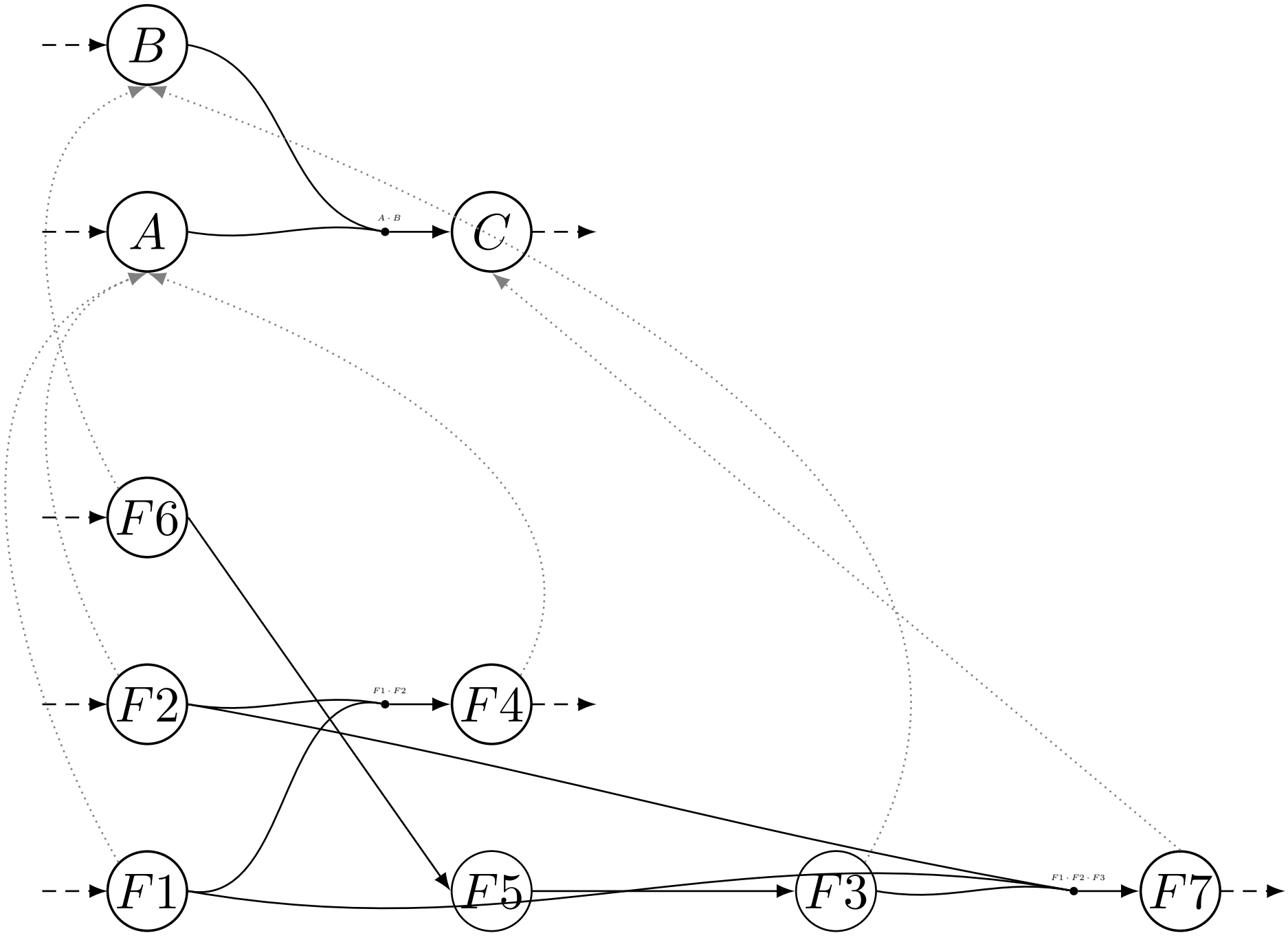
$$(A \cdot B \leftrightarrow C) \cdot (F5 \leftrightarrow F3) \cdot (F1 \cdot F2 \leftrightarrow F4) \cdot (F5 \leftrightarrow F6) \cdot (F1 \cdot F2 \cdot F6 \leftrightarrow F7)$$

solution no. 40/54



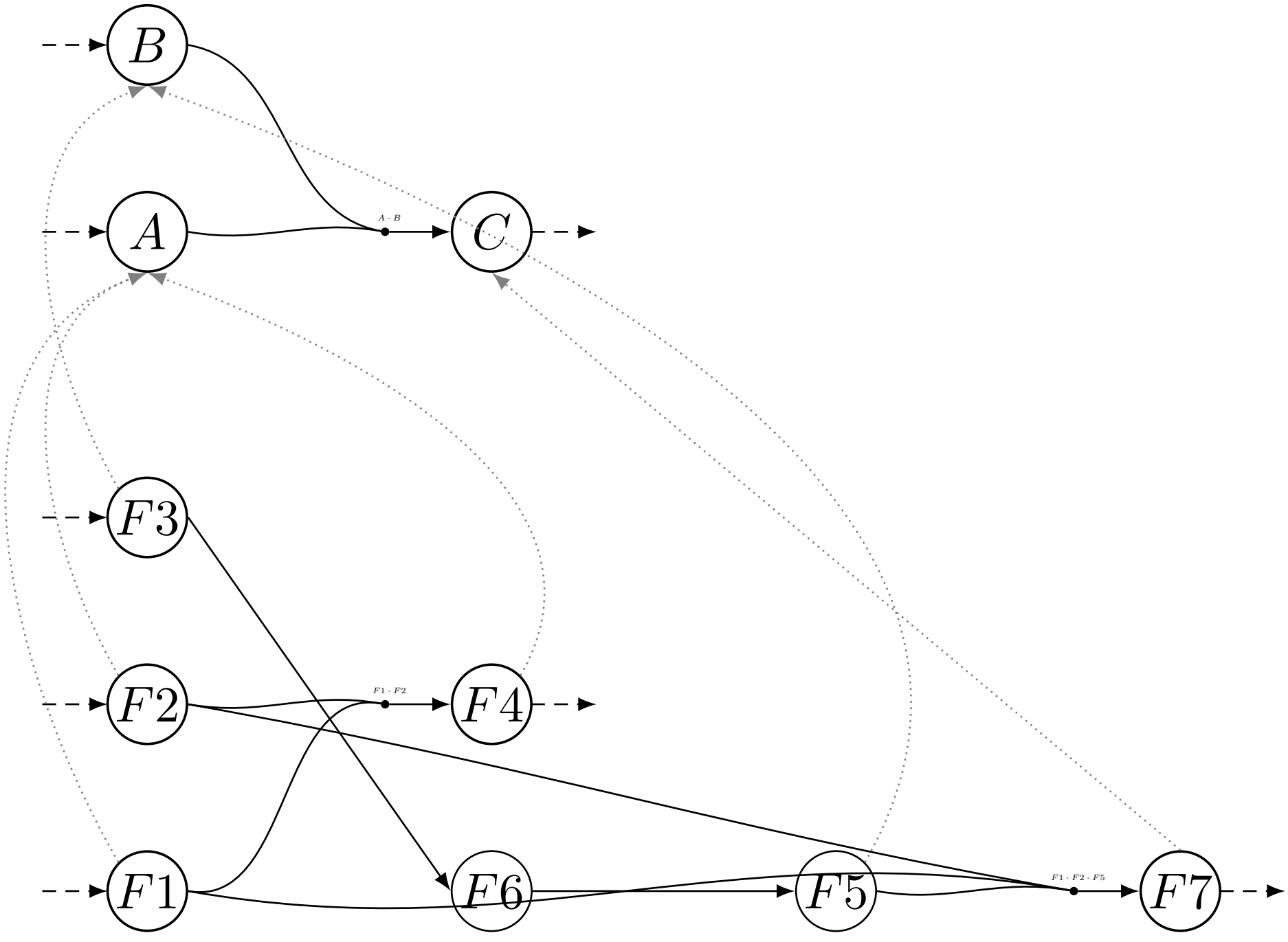
$$(A \cdot B \leftrightarrow C) \cdot (F1 \cdot F2 \leftrightarrow F4) \cdot (F6 \leftrightarrow F5) \cdot (F3 \leftrightarrow F6) \cdot (F1 \cdot F2 \cdot F3 \leftrightarrow F7)$$

solution no. 41/54



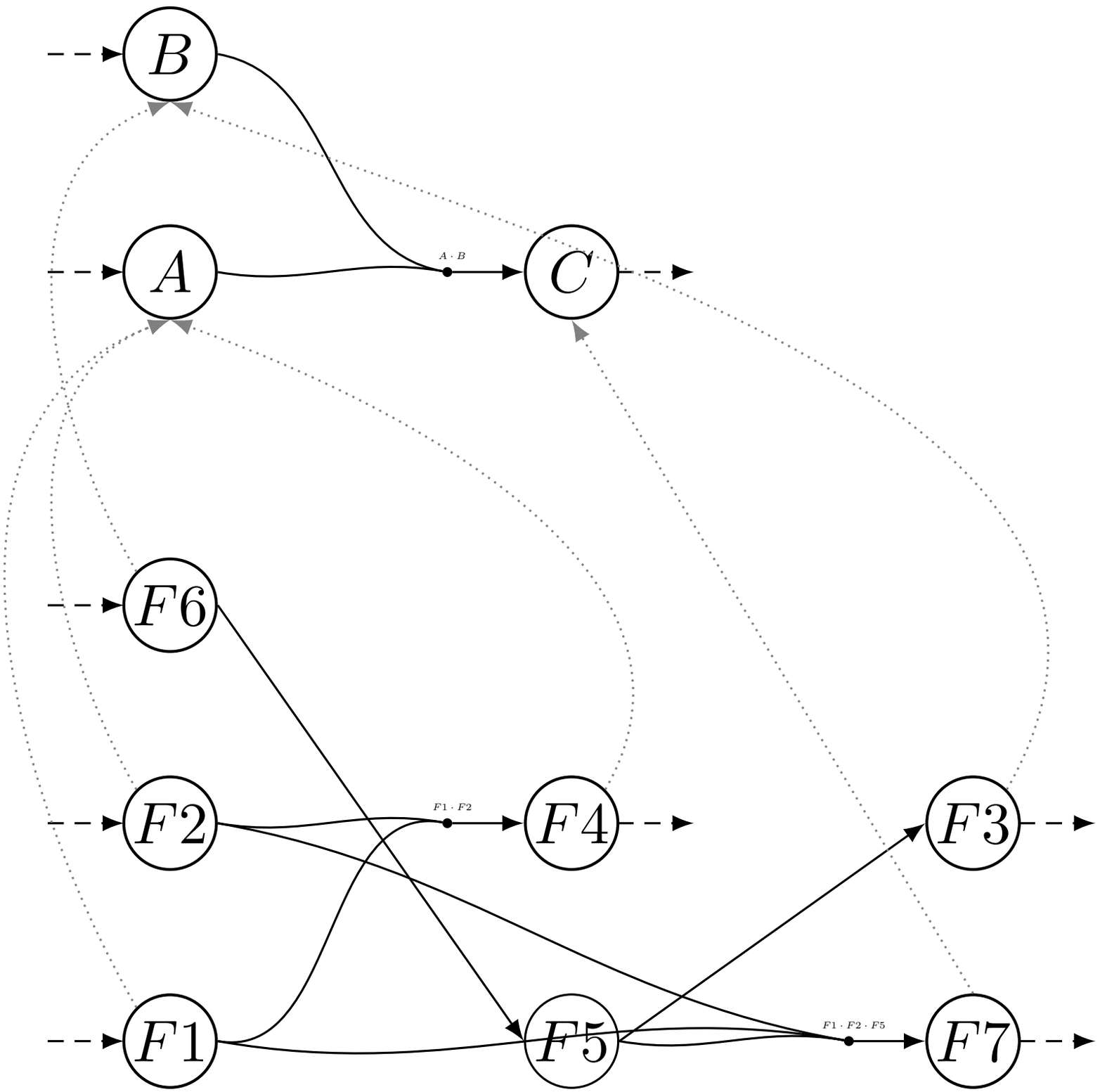
$$(A \cdot B \leftrightarrow C) \cdot (F5 \leftrightarrow F3) \cdot (F1 \cdot F2 \leftrightarrow F4) \cdot (F6 \leftrightarrow F5) \cdot (F1 \cdot F2 \cdot F3 \leftrightarrow F7)$$

solution no. 42/54



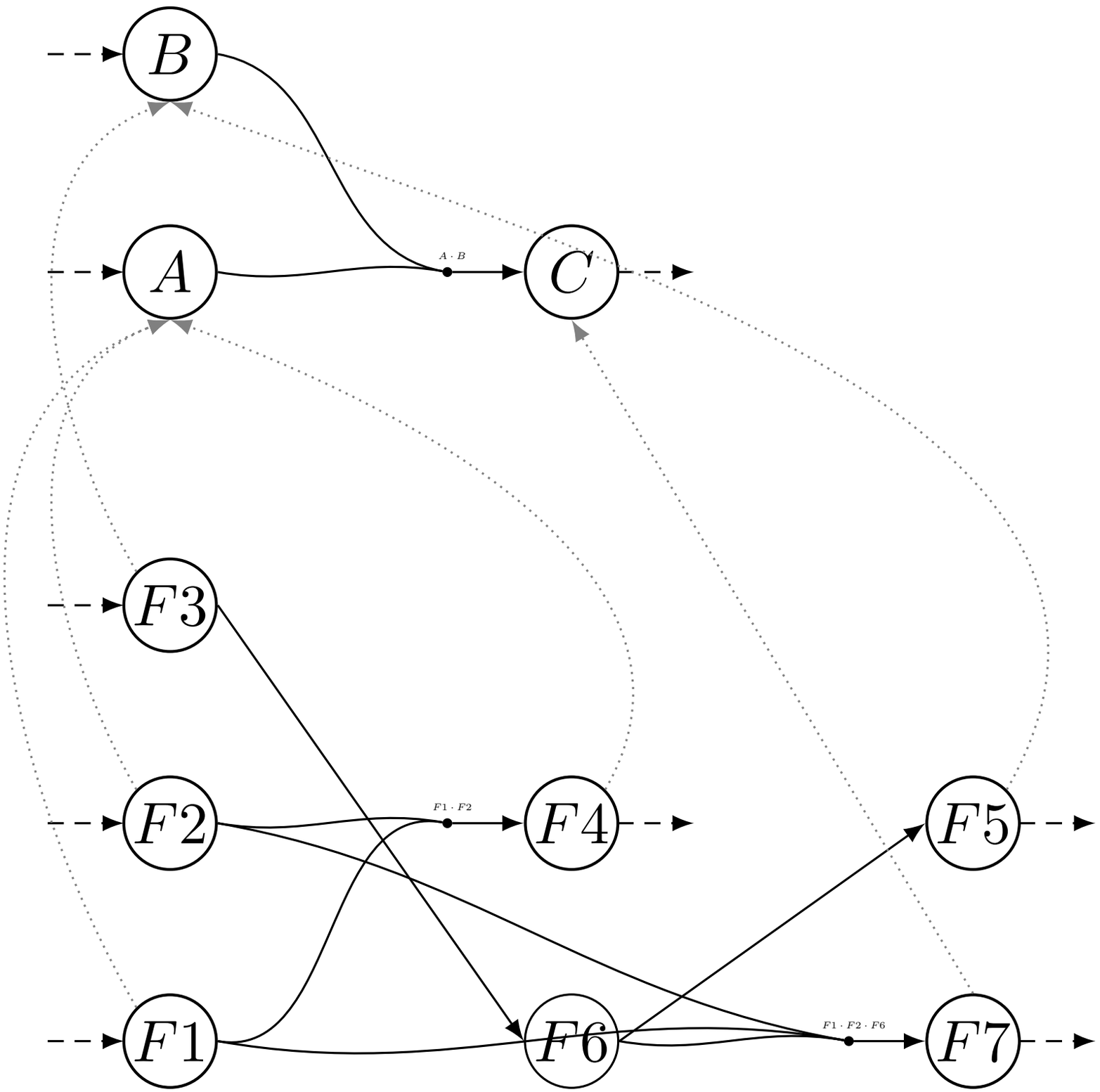
$$(A \cdot B \leftrightarrow C) \cdot (F1 \cdot F2 \leftrightarrow F4) \cdot (F6 \leftrightarrow F5) \cdot (F3 \leftrightarrow F6) \cdot (F1 \cdot F2 \cdot F5 \leftrightarrow F7)$$

solution no. 43/54



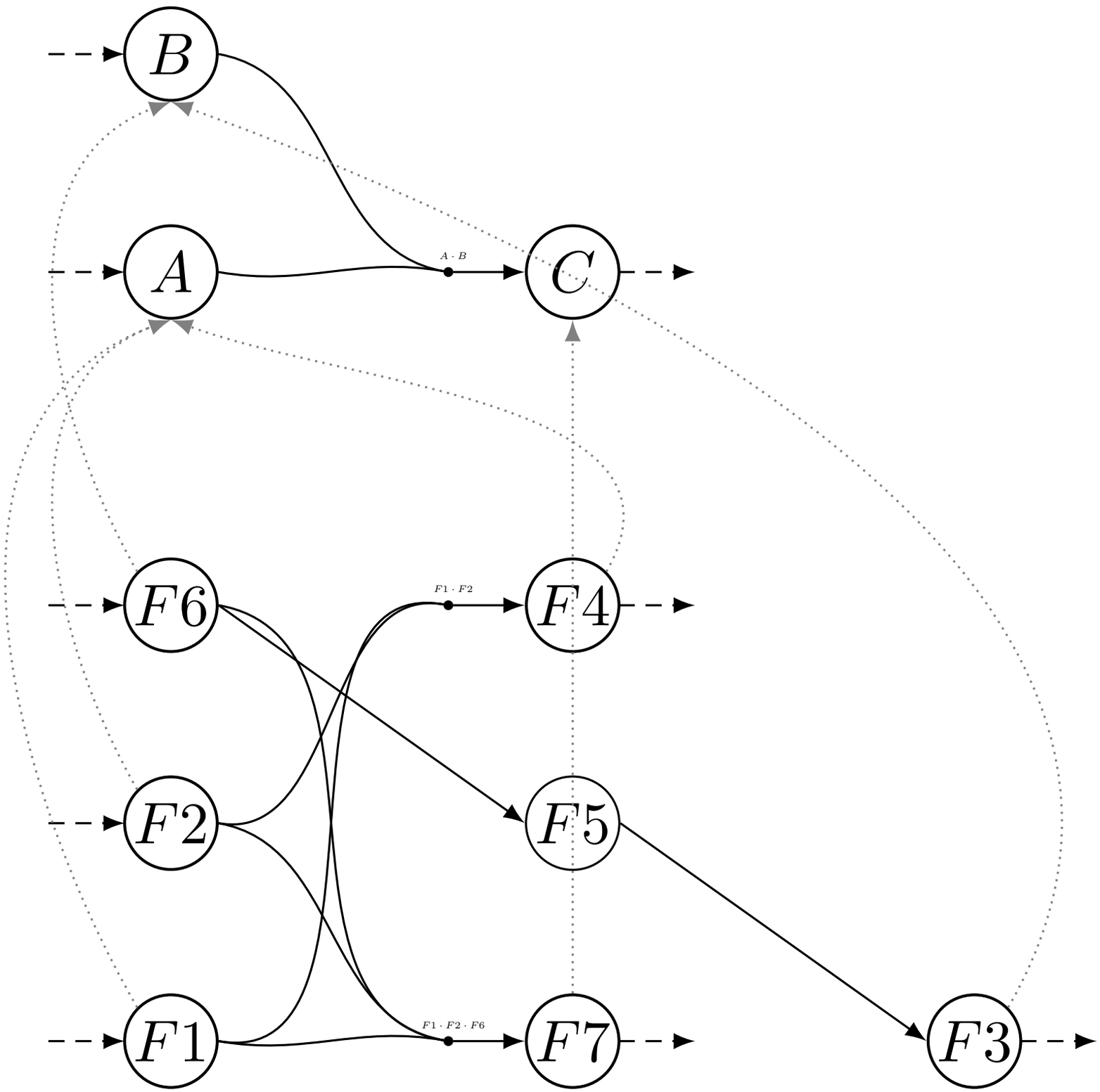
$$(A \cdot B \leftrightarrow C) \cdot (F5 \leftrightarrow F3) \cdot (F1 \cdot F2 \leftrightarrow F4) \cdot (F6 \leftrightarrow F5) \cdot (F1 \cdot F2 \cdot F5 \leftrightarrow F7)$$

solution no. 44/54



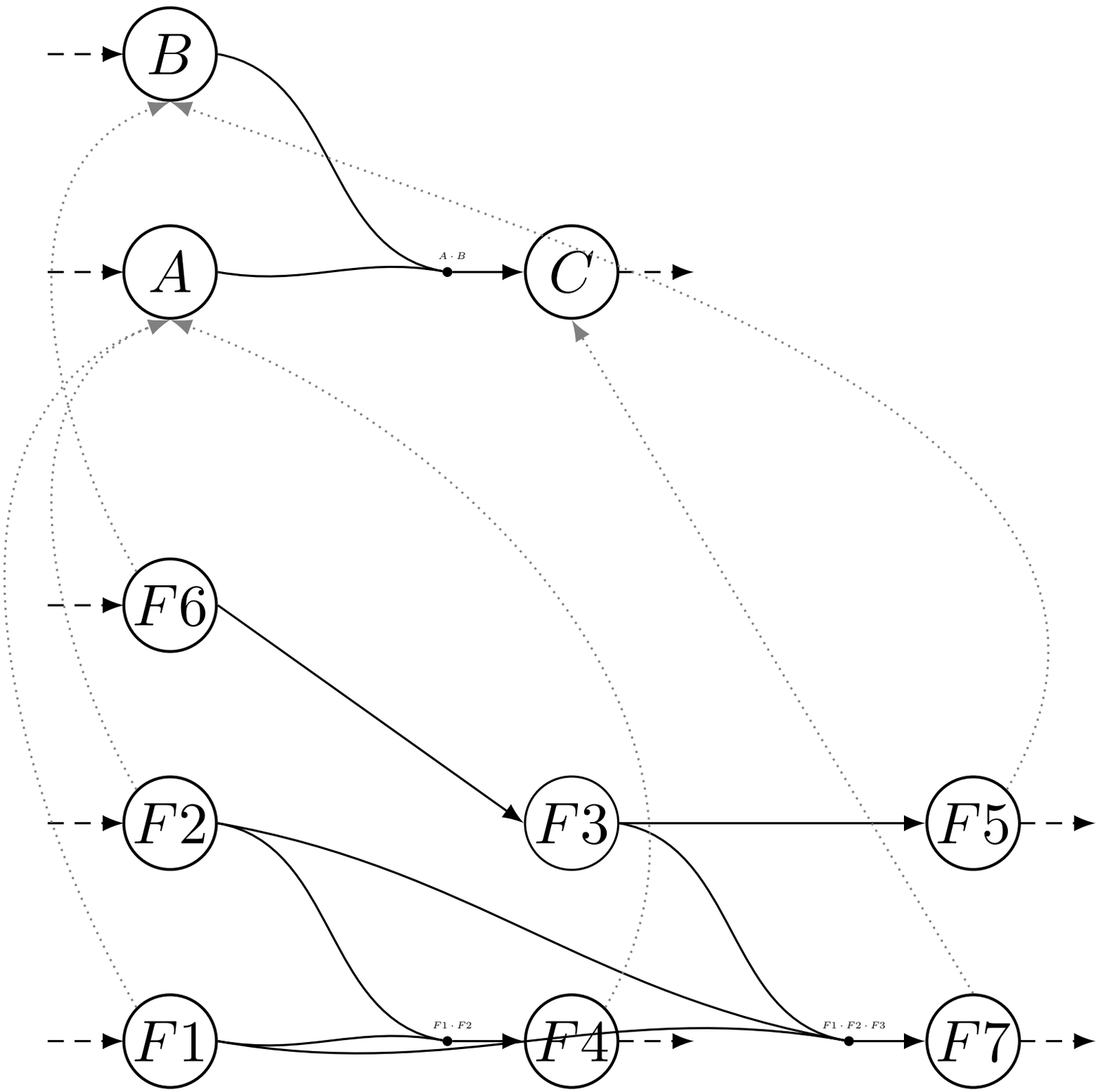
$$(A \cdot B \leftrightarrow C) \cdot (F1 \cdot F2 \leftrightarrow F4) \cdot (F6 \leftrightarrow F5) \cdot (F3 \leftrightarrow F6) \cdot (F1 \cdot F2 \cdot F6 \leftrightarrow F7)$$

solution no. 45/54



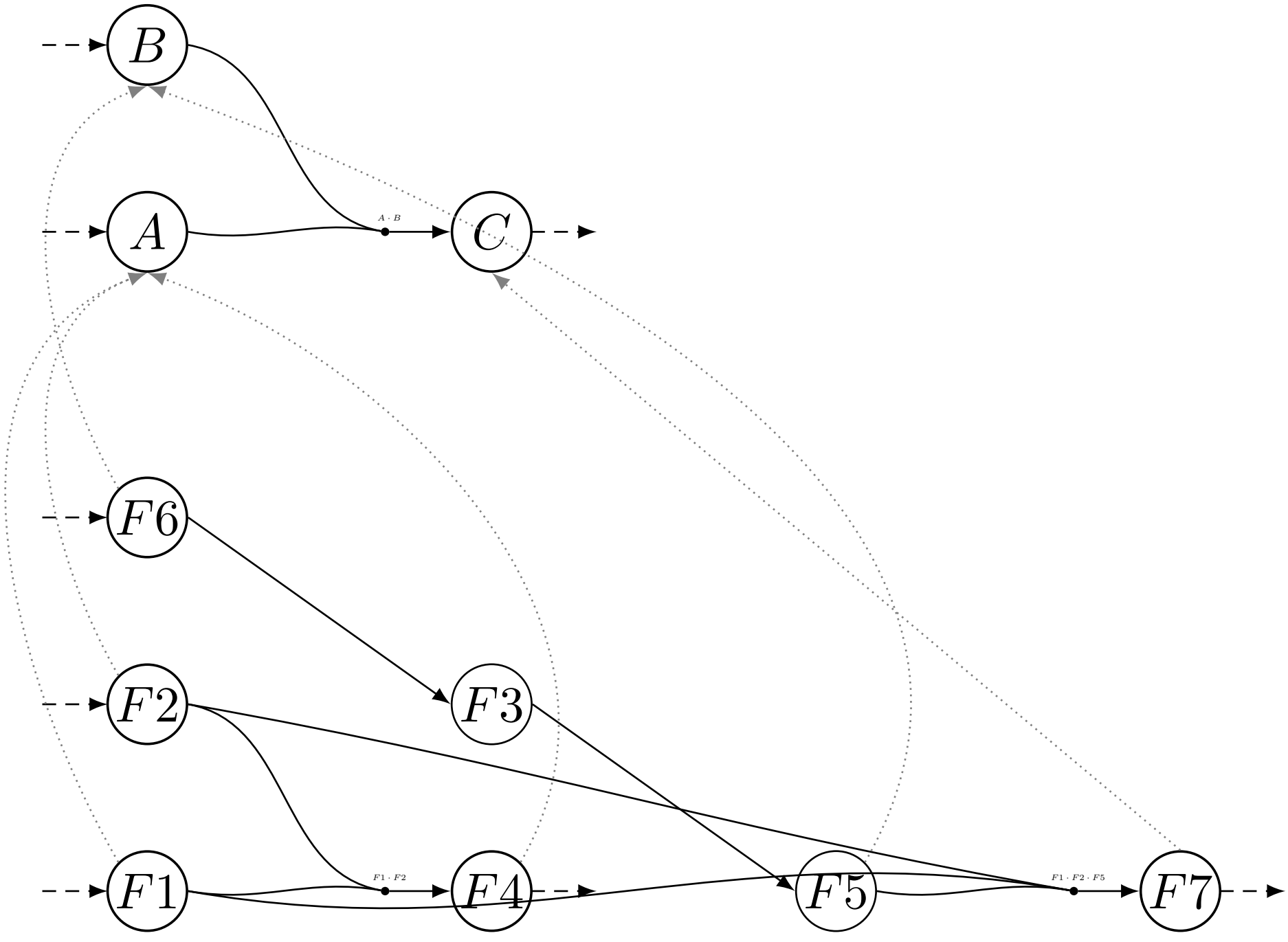
$$(A \cdot B \leftrightarrow C) \cdot (F5 \leftrightarrow F3) \cdot (F1 \cdot F2 \leftrightarrow F4) \cdot (F6 \leftrightarrow F5) \cdot (F1 \cdot F2 \cdot F6 \leftrightarrow F7)$$

solution no. 46/54



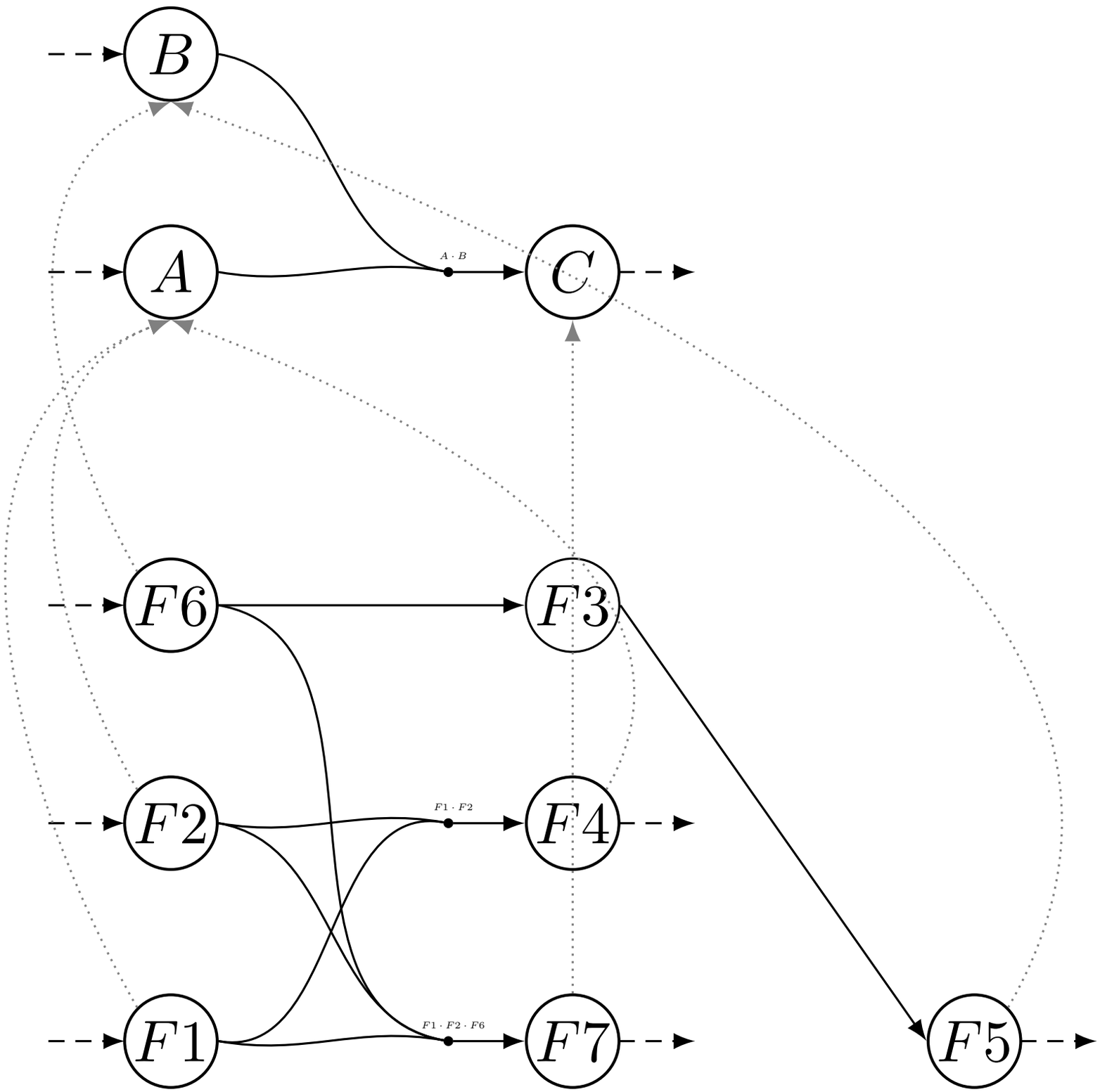
$$(A \cdot B \leftrightarrow C) \cdot (F6 \leftrightarrow F3) \cdot (F1 \cdot F2 \leftrightarrow F4) \cdot (F3 \leftrightarrow F5) \cdot (F1 \cdot F2 \cdot F3 \leftrightarrow F7)$$

solution no. 47/54



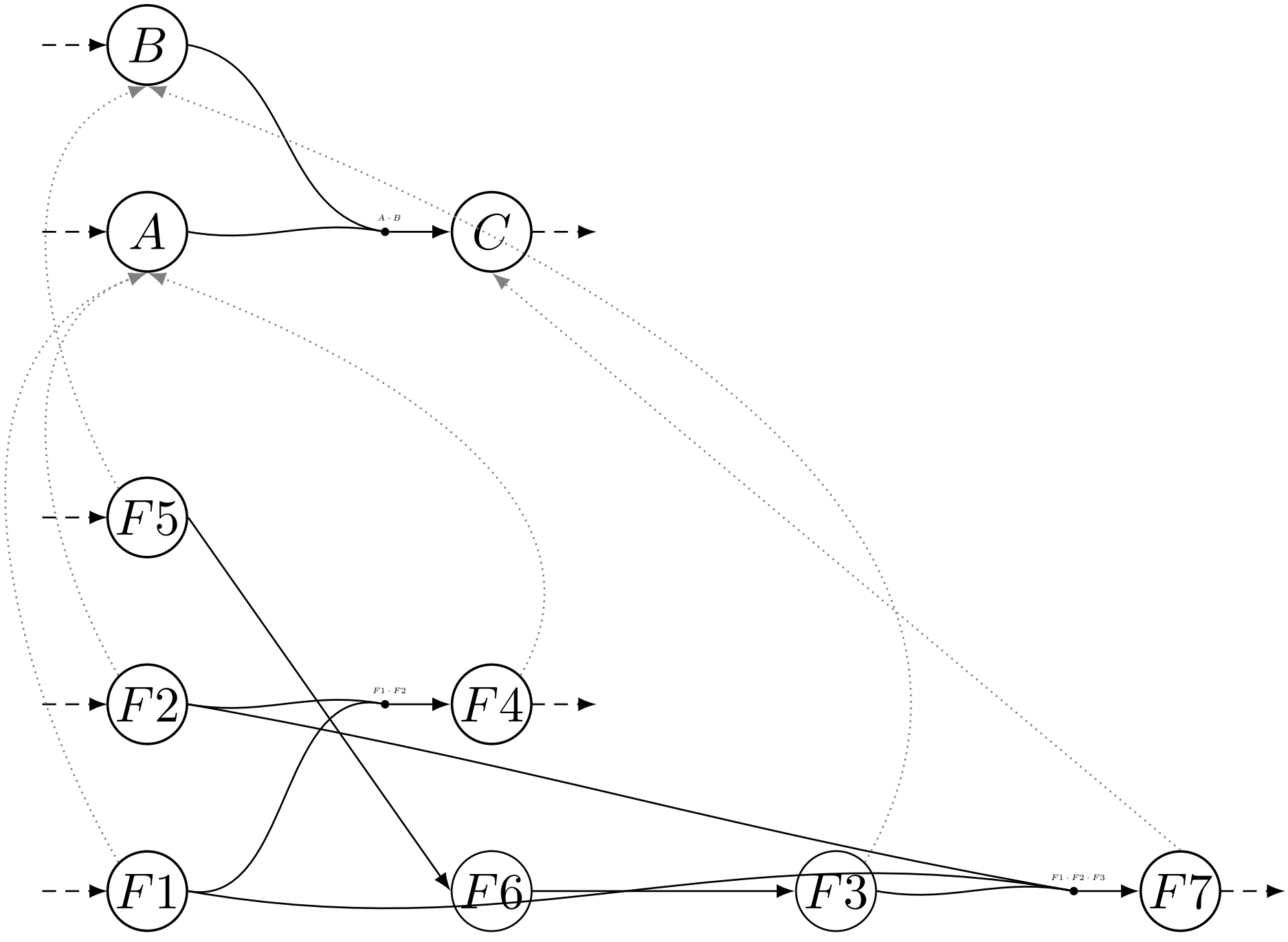
$$(A \cdot B \leftrightarrow C) \cdot (F6 \leftrightarrow F3) \cdot (F1 \cdot F2 \leftrightarrow F4) \cdot (F3 \leftrightarrow F5) \cdot (F1 \cdot F2 \cdot F5 \leftrightarrow F7)$$

solution no. 48/54



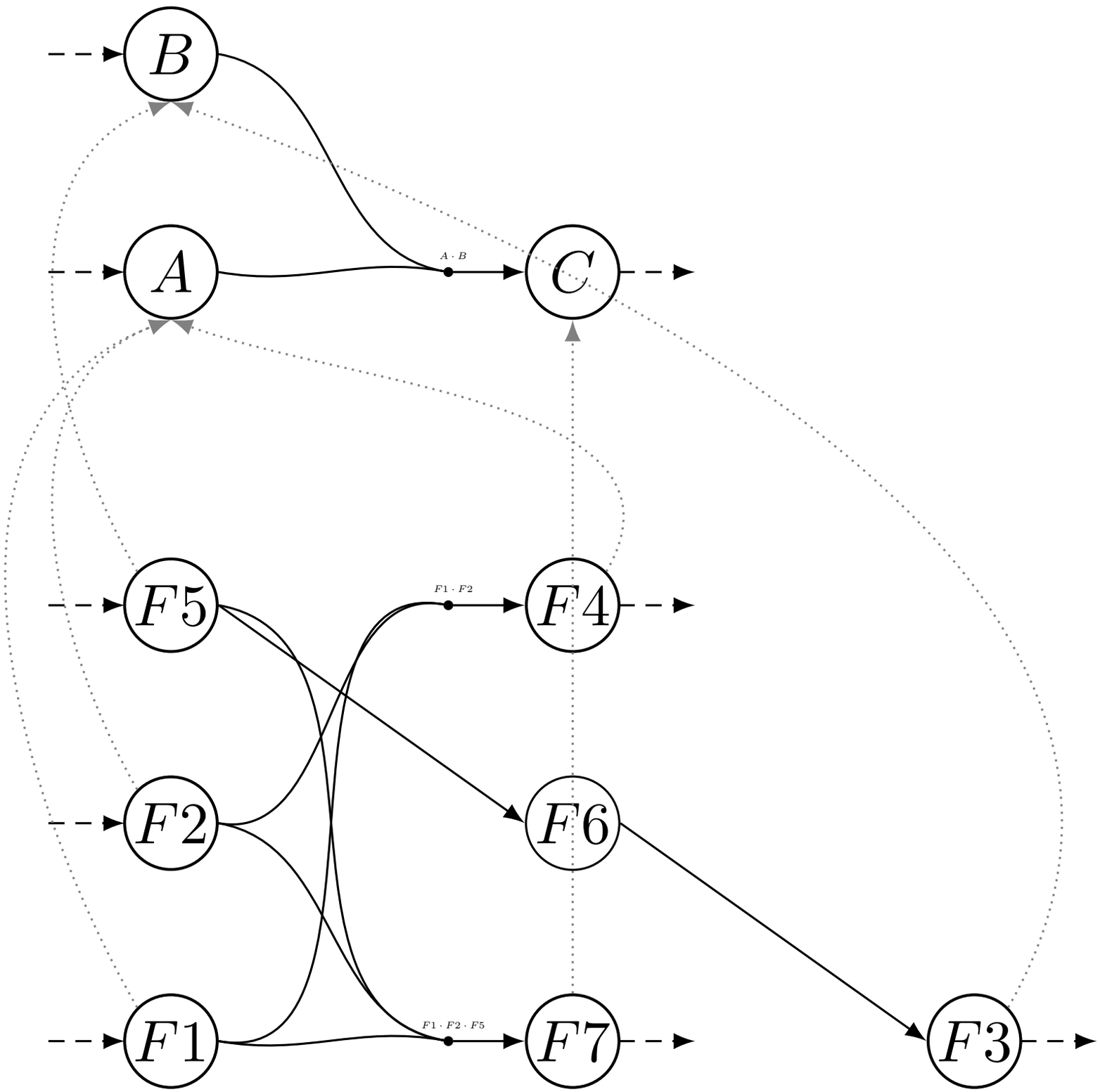
$$(A \cdot B \leftrightarrow C) \cdot (F6 \leftrightarrow F3) \cdot (F1 \cdot F2 \leftrightarrow F4) \cdot (F3 \leftrightarrow F5) \cdot (F1 \cdot F2 \cdot F6 \leftrightarrow F7)$$

solution no. 49/54



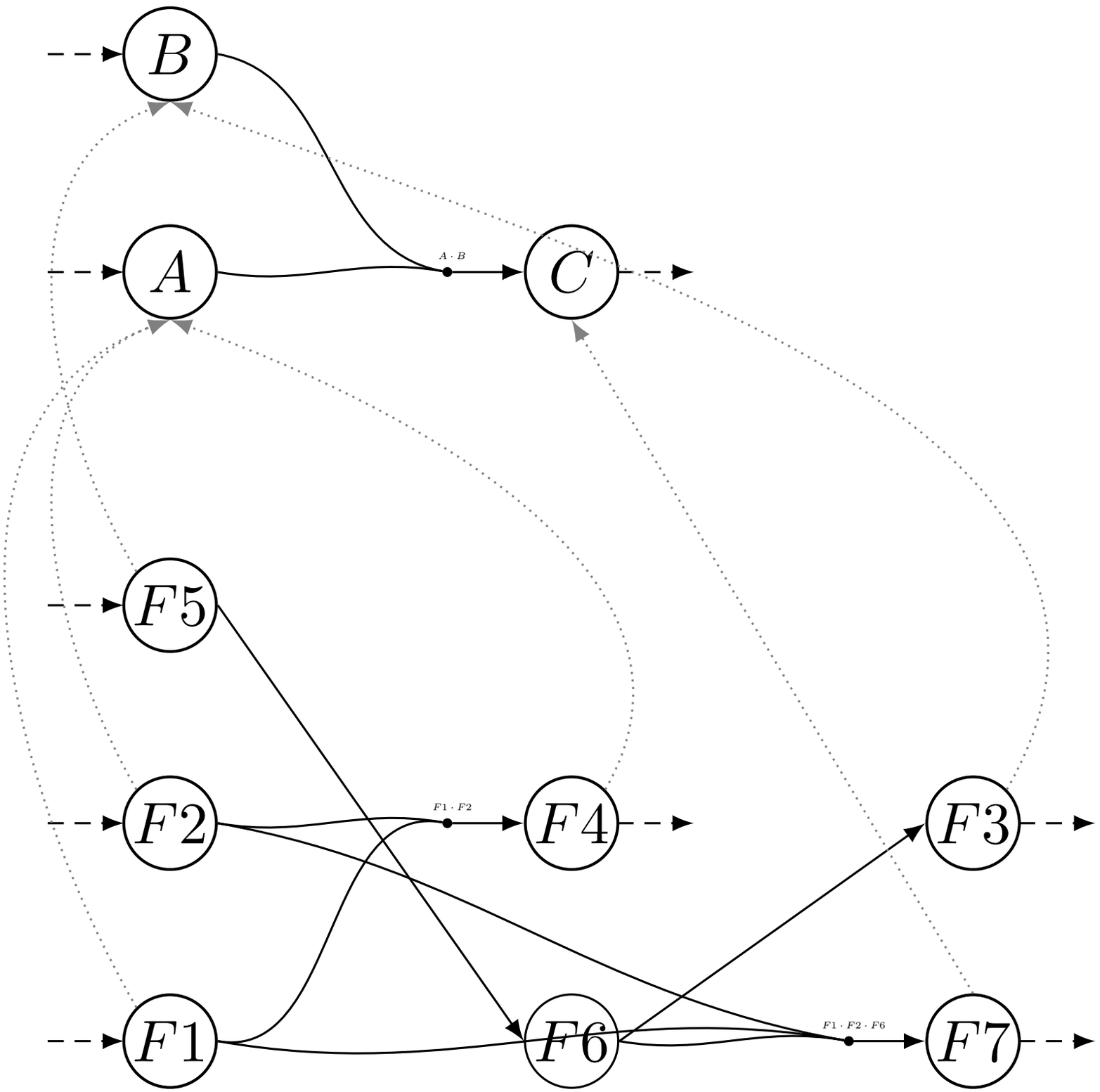
$$(A \cdot B \leftrightarrow C) \cdot (F6 \leftrightarrow F3) \cdot (F1 \cdot F2 \leftrightarrow F4) \cdot (F5 \leftrightarrow F6) \cdot (F1 \cdot F2 \cdot F3 \leftrightarrow F7)$$

solution no. 50/54



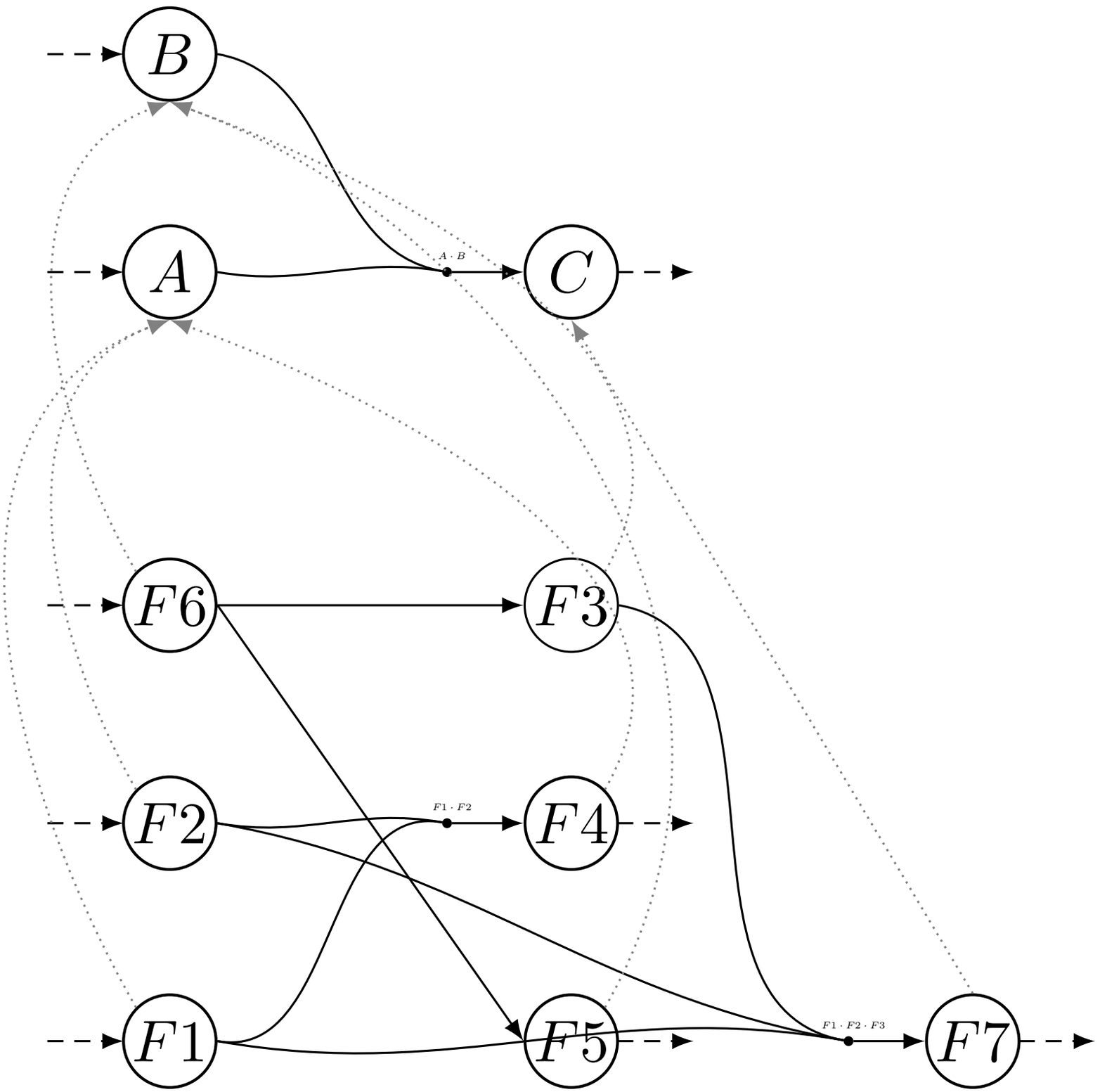
$$(A \cdot B \leftrightarrow C) \cdot (F6 \leftrightarrow F3) \cdot (F1 \cdot F2 \leftrightarrow F4) \cdot (F5 \leftrightarrow F6) \cdot (F1 \cdot F2 \cdot F5 \leftrightarrow F7)$$

solution no. 51/54



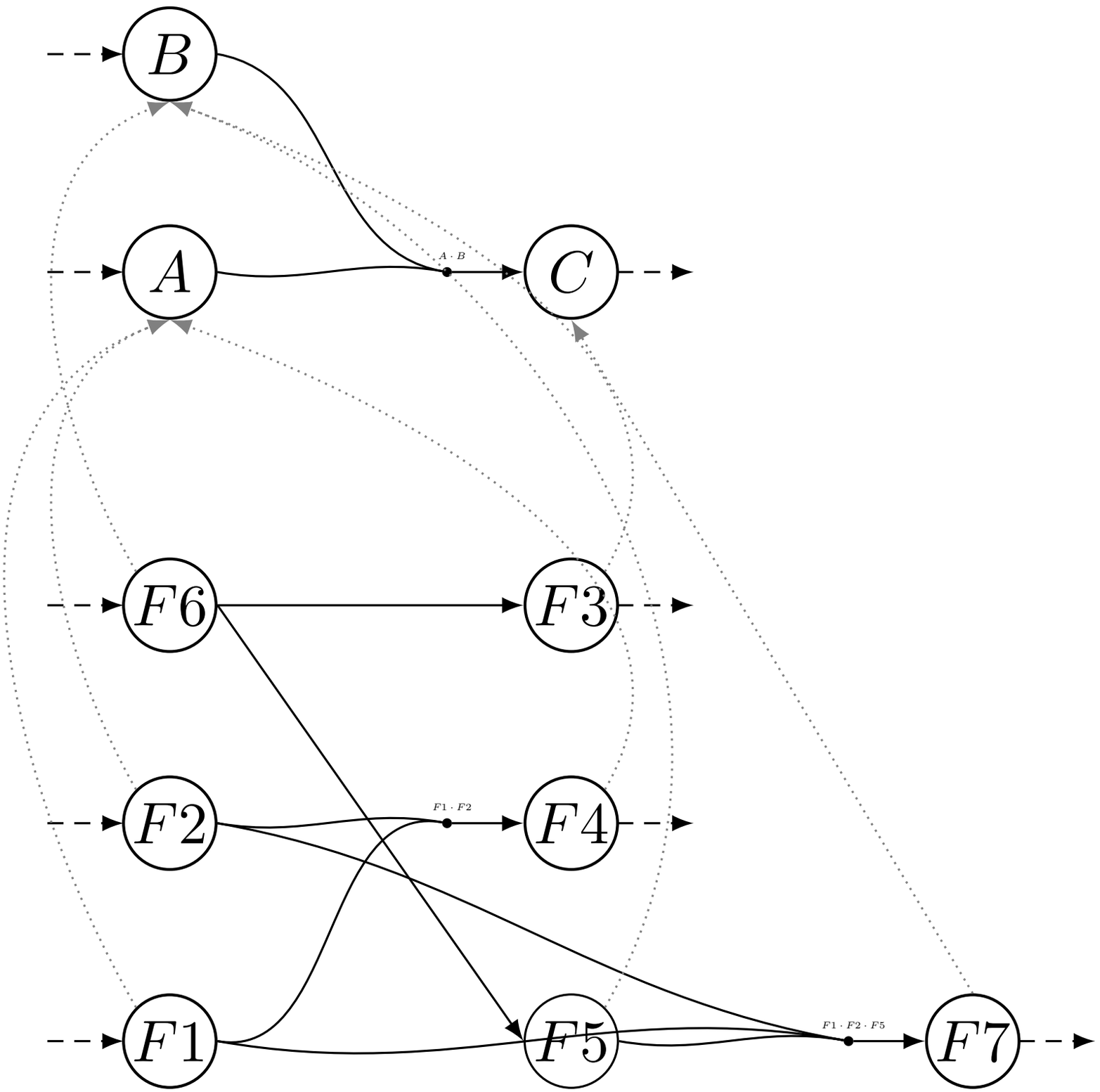
$$(A \cdot B \leftrightarrow C) \cdot (F6 \leftrightarrow F3) \cdot (F1 \cdot F2 \leftrightarrow F4) \cdot (F5 \leftrightarrow F6) \cdot (F1 \cdot F2 \cdot F6 \leftrightarrow F7)$$

solution no. 52/54



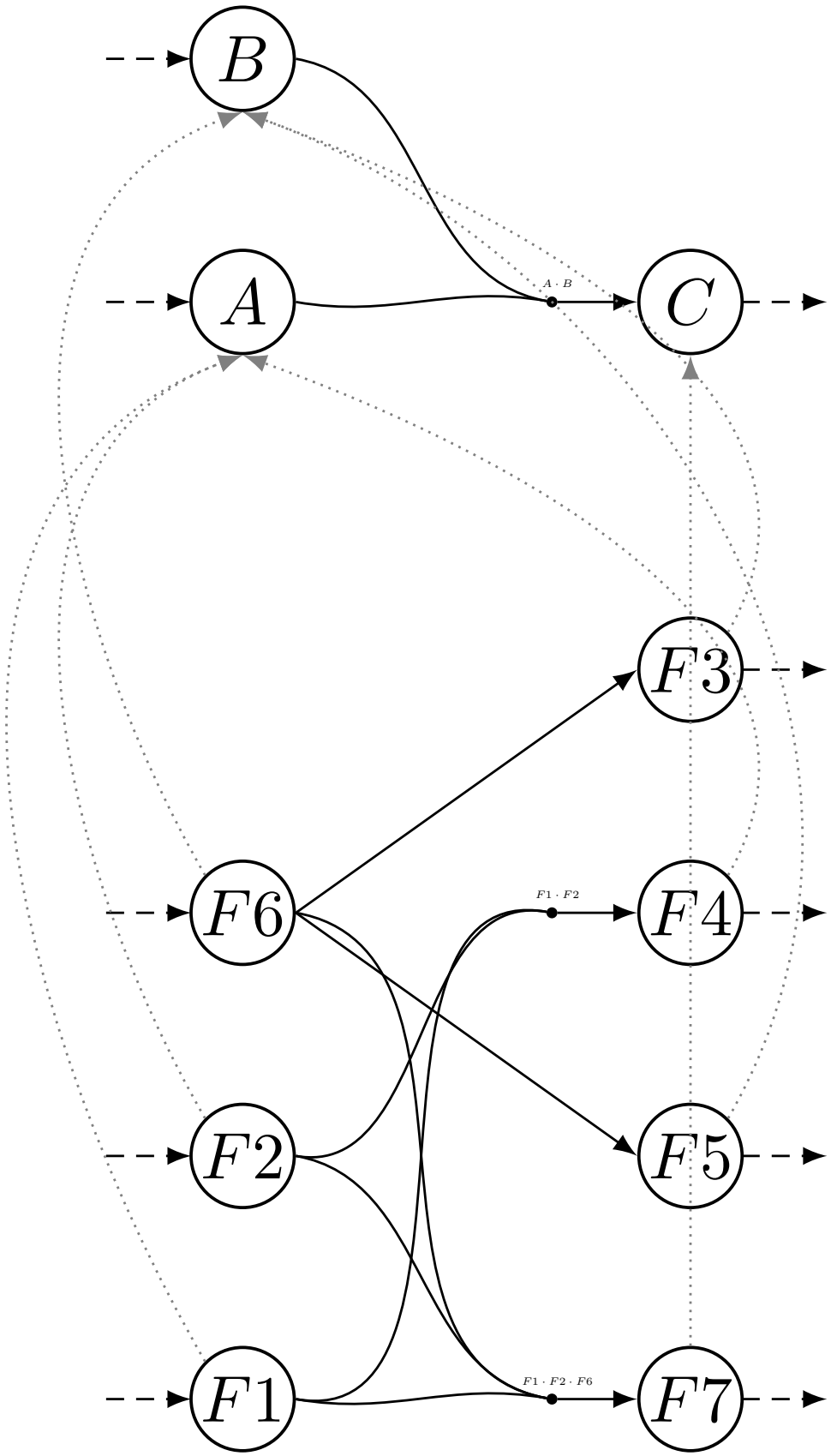
$$(A \cdot B \leftrightarrow C) \cdot (F6 \leftrightarrow F3) \cdot (F1 \cdot F2 \leftrightarrow F4) \cdot (F6 \leftrightarrow F5) \cdot (F1 \cdot F2 \cdot F3 \leftrightarrow F7)$$

solution no. 53/54



$$(A \cdot B \leftrightarrow C) \cdot (F6 \leftrightarrow F3) \cdot (F1 \cdot F2 \leftrightarrow F4) \cdot (F6 \leftrightarrow F5) \cdot (F1 \cdot F2 \cdot F5 \leftrightarrow F7)$$

solution no. 54/54



$$(A \cdot B \leftrightarrow C) \cdot (F6 \leftrightarrow F3) \cdot (F1 \cdot F2 \leftrightarrow F4) \cdot (F6 \leftrightarrow F5) \cdot (F1 \cdot F2 \cdot F6 \leftrightarrow F7)$$